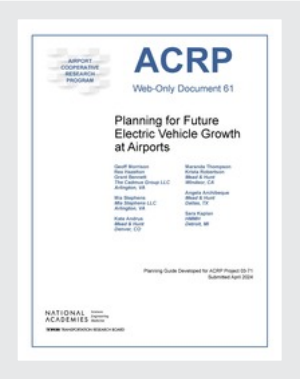
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SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2024. *Planning for Future Electric Vehicle Growth at Airports*. Washington, DC: The National Academies Press. https://doi.org/10.17226/27889.

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AIRPORT COOPERATIVE RESEARCH PROGRAM

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Planning for Future Electric Vehicle Growth at Airports

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Planning Guide Developed for ACRP Project 03-71
Submitted April 2024

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ACRP

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ACKNOWLEDGMENT

This work was sponsored by the Federal Aviation Administration (FAA). It was conducted through the Airport Cooperative Research Program (ACRP), which is administered by the Transportation Research Board (TRB) of the National Academies of Sciences, Engineering, and Medicine.

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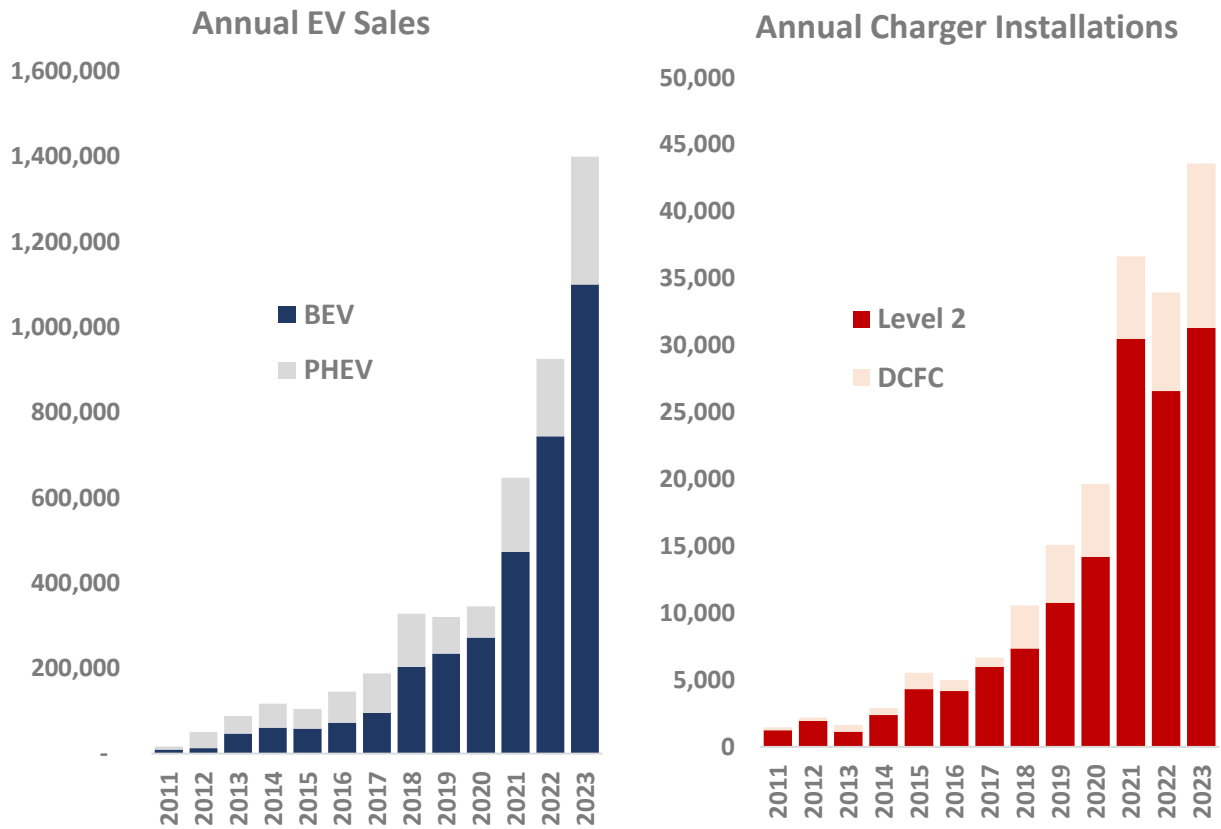
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Summary

As electric vehicle (EV) populations increase across the United States, airports have an elevated role in planning for charging infrastructure. Annual EV sales have increased dramatically over the past decade; over 900,000 EVs were sold in 2022 in the United States, accounting for 8% of all new light-duty vehicle (LDV).[1] Figure 1 shows historical EV sales and charger installations.

As of July 2023, nearly four million EVs and 160,000 public electric vehicle service equipment (EVSE) ports were in use in the United States.[2] Going forward, this growth is expected to continue. The White House has established a goal of 50% of new LDV sales to be electric by 2030.[3] The U.S. Department of Energy (DOE) estimates that by 2030 there will be 33 million EVs on the road and 10 times more public chargers will be needed than in 2023.[4]

Figure 1. Annual U.S. Electric Vehicle Sales and Public Charger Installations



Source: Alliance for Automotive Innovation (left), U.S. DOE Alternative Fuels Data Center (right)

This transformation is already impacting landside and airside operations at airports. Electric ground support equipment (eGSE), passenger vehicles, rental car fleets, buses, and other vehicle types are being deployed in increasingly greater numbers. However, the rate of EV deployment differs by region. In California, Washington, Oregon, Nevada, and Colorado, EVs accounted for more than 10% of new passenger vehicle sales in 2022 and the first quarter of 2023.[5] On the other hand, the six states with the

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lowest EV penetration had EV sales rates of less than 2%. Much of the difference at the state level is driven by state-level policy. For example, 16 states representing roughly one-third of the U.S. vehicle market have adopted regulations to achieve 100% zero-emissions vehicle (ZEV) sales for passenger vehicles by 2035.[6] Furthermore, other states have regulations requiring electrification of medium- and heavy-duty (MDHD) fleet vehicles, such as transit buses, delivery vehicles, and ground support equipment. For example, in California the *Zero-Emission Airport Shuttle Regulation* requires 100% of airport shuttle operators at the 13 largest airports in the state to transition to 100% ZEVs by December 31, 2035, and all new shuttle vehicles purchased after 2023 must be ZEVs.[7]

As the number of EVs at airports scales, so does the demand for EV chargers—and the demand for electricity. One recent estimate suggests that electricity demand at airports may double within ten years or less.[8] Certain charger types in particular—such as eGSE—create large loads which can require extensive upgrades to the distribution system (Figure 2). The new electricity demand is also an opportunity for airports. For example, EVs provide airports with new ways to generate revenue from tenants, passengers, ground transportation organizations, and others.

Because of the diversity of charging needs at airports and the varied policy environments in which airports operate, there is no single roadmap that describes how to best install EV charging. Further, an airport may use multiple approaches to owning and operating charging infrastructure. To help simplify this complex space, this guide categorizes charging into **eight use cases** (Figure 3), including eGSE, electric aircraft, passenger and employee parking, airport fleet vehicles, non-airport owned airside vehicles, bus and shuttlebus, rental cars, and for hire vehicles. To the extent possible, the guide points out lessons and resources specific to each use case. However, much of the information is not use case-specific.

This guide was developed as a primer to help educate airport staff and practitioners on the technologies and processes associated with charging infrastructure. Using a question-and-answer format, the guide walks readers through concepts like charging station ownership, load shapes, fees, siting, metrics for tracking charging, and more. The guide has been designed to be useful to readers with little to no familiarity with EV planning.

Figure 2. eGSE Chargers at San Diego International Airport



Source: Cadmus

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Figure 3. Eight Airport Use Cases Defined in this Guide.

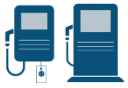
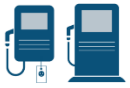
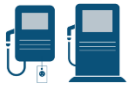
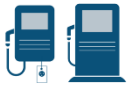
Source: Consultant Team

Use Cases are vehicle-charger combinations, defined by their unique operational and charging characteristics (e.g., duty cycle, power level, plug type, time of use, load curve).

Level 1 <1.9 kW 

Level 2 <19.2 kW 

DCFC 50kW+ 

Use Case	 Passengers, Employees, Car Share	 Rental Cars	 For Hire Vehicles	 Bus & Shuttlebus	 Airport Fleet Vehicles	 Non-Airport Owned Airside Vehicles	 Ground Support Equipment	 Electric Aircraft
Location	Landside	Landside	Landside	Airside and Landside	Airside and Landside	Airside	Airside	Airside
Access	Public	Public and Restricted	Public	Public and Restricted	Public and Restricted	Restricted	Restricted	Restricted
Dwell Time	Hours to Days	Minutes to Hours	Minutes to Hours	Minutes to Hours	Minutes, Hours or Days	Minutes, Hours or Days	Minutes to Hours	Minutes to Hours
Charger Owner	Airport, Third Party	Airport, RAC, Third Party	Airport, TNC, Taxi, Third Party	Airport, Third Party	Airport, Third Party	Airport, Tenant, Third Party	Airport, Airline, FBO, Third Party	Airline, FBO, Third Party
Charging Port Type	J1772, J3400, CCS	J1772, J3400, CCS	J3400, CCS	J1772, J3400, CCS	J1772, J3400, CCS	J1772, J3400, CCS, Proprietary	J1772, J3400, CCS, Proprietary	CCS, MCS, Proprietary
Charging Power Level								

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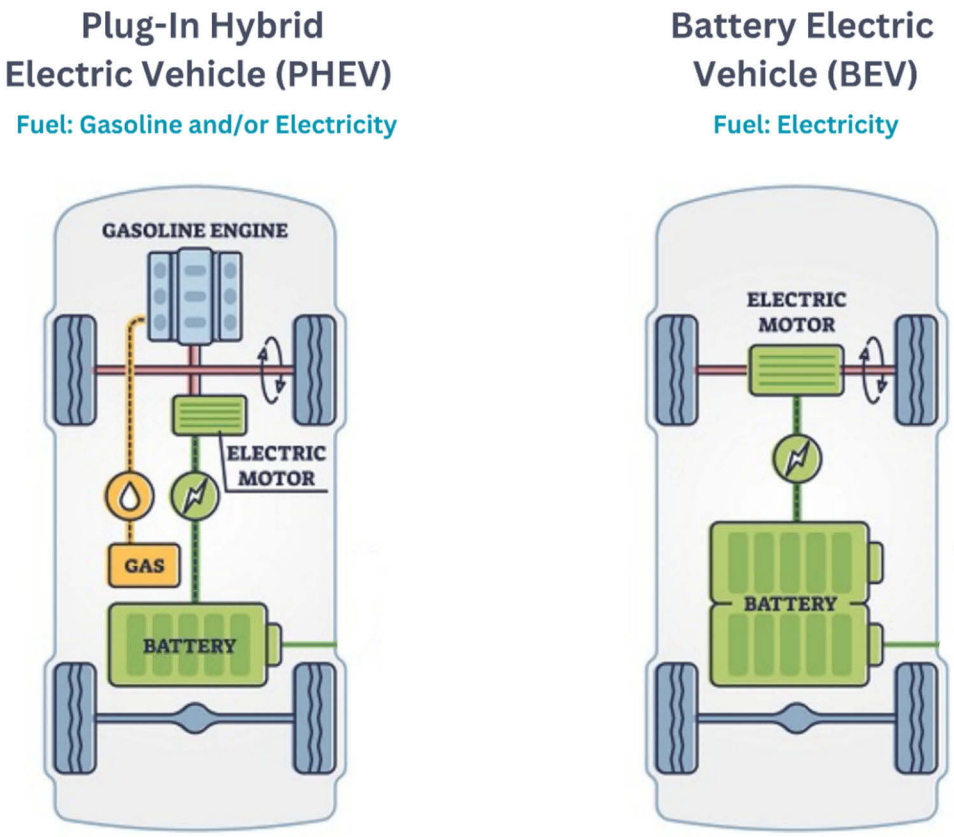
Chapter 1. Fundamental Topics on EV Charging

This chapter provides foundational concepts related to EVs and charging infrastructure. While the information presented in this chapter is primarily focused on light-, medium-, and heavy-duty vehicles, most information is applicable to other types of EVs including specialty vehicles and aircraft. Chapter 2 provides information on more advanced topics.

What are electric vehicles?

EVs use electrical energy, stored in batteries inside the vehicle, for propulsion via an electric motor. There are two primary types of EV powertrains (Figure 4): plug-in hybrid electric vehicles (PHEVs) and battery electric vehicles (BEVs). Collectively these two vehicle types are referred to as EVs.

Figure 4. Electric Vehicle Power and Fueling Systems



Source: Consultant Team & United States Department of Transportation

- **PHEVs** are powered by both an internal combustion engine (ICE) and an electric motor with a battery. Unlike conventional hybrid vehicles, PHEVs can recharge their batteries using an EVSE port.[9]
- **BEVs** are powered by an electric motor and use a battery that is recharged using an EVSE port and electrical outlet or hardwired EVSE.[10]

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Today's PHEVs have an all-electric range of 20 miles to over 100 miles. There are currently 33 light-duty PHEV models available in the United States. BEVs have a range of 80 miles to more than 500 miles, depending on the model. There are currently 40 BEV models available in the United States, and this number is expected to increase to over 100 by 2025.[11]

Between 2011 and 2023, sales of EVs rose from 0.2% to 8% of new LDVs nationally. A 2021 federal Executive Order has a target of 50% of U.S. passenger car and light truck sales as ZEVs by 2030.[12] Vehicle and battery manufacturers plan to invest $860 billion globally in EVs between 2023 and 2030, with $210 billion targeted for the United States.[13] Through the Infrastructure Investment and Jobs Act (IIJA) and Inflation Reduction Act (IRA), the U.S. government has made upward of $83 billion in funding available to support the roll out of EVs. Moreover. the average price of a light-duty EV fell by 20% between 2022 and 2023, down to $53,000 compared with $48,000 for an ICE vehicle,[14] and average EV range has increased from 84 miles to 275 miles in the past decade.[15]

While LDVs are the most common EVs available today, there are applications across all vehicle segments, including both on-road and off-road vehicles (Figure 5). In recent years, electric school and transit buses have begun to achieve widespread adoption.

Figure 5. Examples of Medium- and Heavy-Duty Electric Vehicles Available Today



Source: Consultant Team

What are the elements of charging infrastructure?

Charging infrastructure is a broad term and typically refers to the structures and equipment necessary to support an EV, including the connector, cable, pedestal, wiring, conduits, substations, and transformers that provide electricity to the vehicle. Electric utilities often define charging infrastructure in two parts: to-the-meter **(TTM) infrastructure** and **behind-the-meter (BTM) infrastructure** (Figure 6).

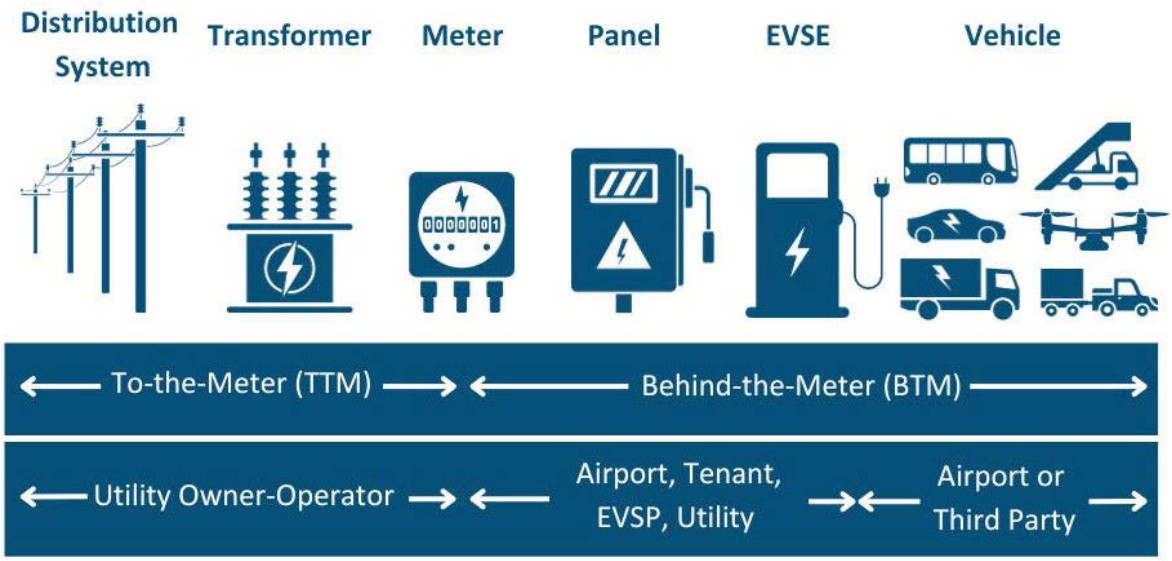
As shown in the bottom row of Figure 6, the electric utility owns and operates infrastructure to the meter. This means upgrades for TTM infrastructure require coordination with the electric utility and in some cases co-payment by the installer of the EVSE port. However, in a growing number of regions, costs for upgrades of TTM are born solely by the electric utility.[16] The BTM infrastructure is all structures and equipment between the electric meter and EVSE port. Ownership, operation, and funding for BTM infrastructure varies and can include the site host, tenant, third party, or utility.

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Electric Vehicle Supply Equipment (EVSE) comprises the electrical conductors and equipment external to an EV that provides a connection between an EV and a power source to enable charging. Because it is only the hardware and enclosure to bring power from the grid to the vehicle (like a gas pump for a vehicle with an ICE), EVSE is only one component of a charging infrastructure project.

An **Electric Vehicle Service Provider (EVSP)**, also called a **Network Service Provider (NSP)** is an entity responsible for operating networked or non-networked EVSE, which may include communications systems, billing, maintenance, load management, and other processes. Examples of EVSPs or NSPs in the United States are ChargePoint, Electrify America, and EVgo.

Figure 6. Elements of Charging Infrastructure. The Bottom Row is the Owner-Operator.



Source: Consultant Team

The Federal Highway Administration and Department of Energy have defined additional infrastructure terminology that aligns with the Open Charge Point Interface (OCPI) adopted by the charging infrastructure industry to describe a charging site:[17] These components are shown in Figure 7.

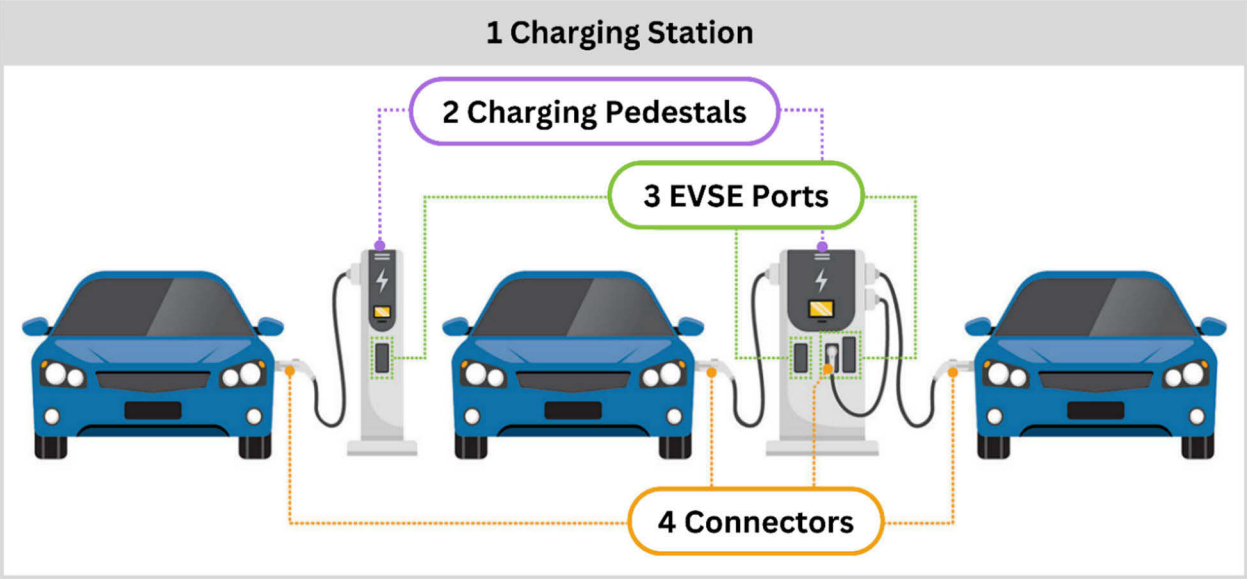
- **Charging station (or station location):** Site with one or more EVSE ports at the same address. Charging stations can be found in parking garages, highway rest areas, parking lots and other sites.
- **Charging pedestal (also called a charging post or charging point):** The unit that houses all equipment such as EVSE ports, user display screen, payment system, and electronic hardware. A pedestal with two EVSE ports can charge two vehicles at the same time. Charging pedestals are sometimes referred to as EVSE.
- **EVSE port:** Unit that provides power to charge one vehicle at a time. The number of ports on a charging pedestal is equal to the number of vehicles that can concurrently charge at that pedestal. This piece of infrastructure is commonly referred to as a charger.

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- **Connector:** Physical attachment at the end of the cable that links the EVSE to the vehicle. Multiple connectors and connector types (e.g., J1772, CCS, CHAdeMO) can be available on a charging pedestal to serve different EV types.

Like a gas station dispenser, only one EVSE port on a given charging pedestal can be used at a time. Importantly, when government websites and documents report statistics like "number of ports," they are referring to EVSE ports, so a two-connector pedestal would count as a single port, but a dual port pedestal would count as two. Public charging station locations, EVSE ports and EVSE characteristics (such as power rating) are tracked by data aggregators like the U.S. DOE Alternative Fuels Data Center[18] and Plugshare.com.[19] Data on private EVSE installations is not available at a national level.

Figure 7. Select Elements of BTM Charging Infrastructure



Source: Alternative Fuels Data Center & Cadmus

What types of connectors are used for EV charging?

EV charging connector types have evolved over the past decade, and there is currently no national standard for the connectors used for charging EVs. However, all EVs in the United States can use the standard Society of Automotive Engineers **(SAE) J1772** connector shown in Figure 8. This type of connector is used only for Level 1 and Level 2 charging with AC power and is the most common connector for most EV chargers today. Tesla vehicles require an adapter to use this kind of connector.

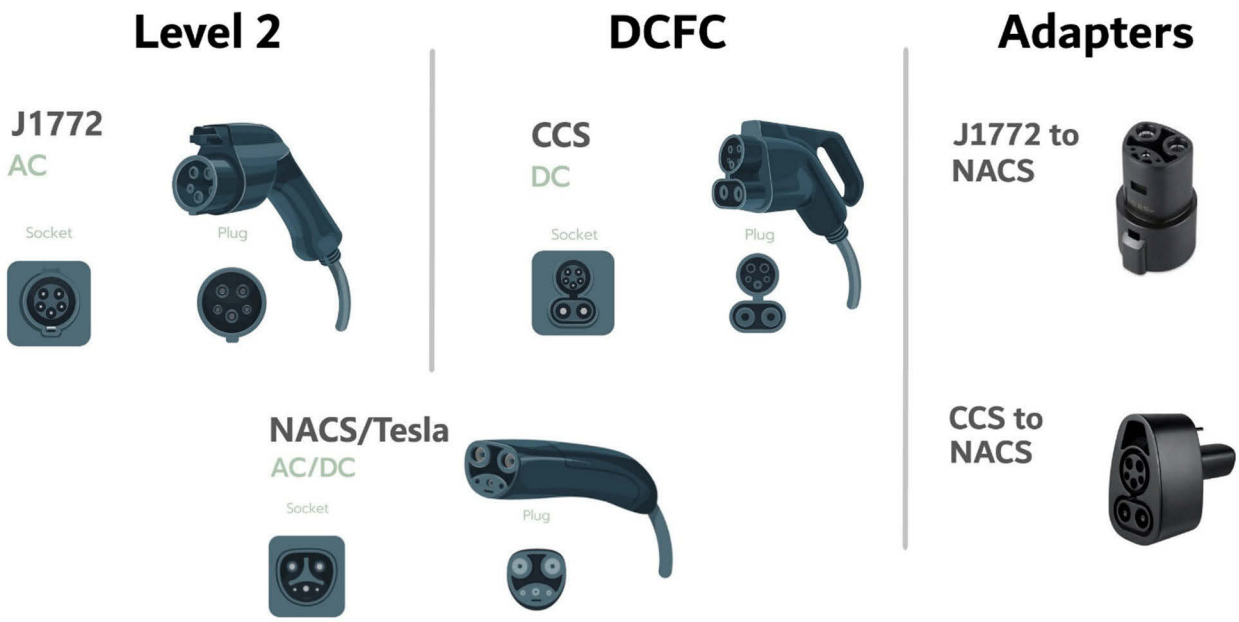
The standardization of connectors is an ongoing topic for DCFC. There are three types of connectors used today. **CHAdeMo** was the first type of DCFC connector developed and is used by Japanese- and Korean-made EVs. While this type of connector can still be found at many public DCFC stations, it is no longer used for new EVs in the United States.

The **Combined Charging System (CSS)**, shown in Figure 8, accounts for roughly one-third of DCFC EVSE ports in the United States. The CCS connector uses the SAE J1772 connector and incorporates two high-speed DCFC pins. The majority of EVs in the United States today have a connector that is compatible

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with the CCS connector. Currently federal regulations require all DCFC stations constructed using federal funds to include the CCS connector.

Figure 8. EVSE Connector Types



Source: Consultant Team

At the time of this writing, all federally funded DCFC charging sites are required to have the CCS connector, while the NACS connector can also be offered but is not required. In June 2023, SAE announced intentions to standardize the NACS connector.[20] It is likely that both L2 and DCFC chargers will continue to use both CCS and NACS connectors and that the LDV market will shift to the NACS standard in the coming years, though it is unclear which connector type MDHD Original Equipment Manufacturers (OEMs) will adopt.

Another type of connector that may be desirable for airports in the future is the **Megawatt Charging Standard (MCS)**, which is intended for charging vehicles with very large batteries, such as long-haul freight trucks and aircraft (Figure 9). These vehicle types can require faster charging speeds and power delivery than are available with CCS or CHAdeMO. The MCS is in development and may be commercially available in 2024. It is modeled on the CCS connector and is expected to charge up to 1,250V.

Many aircraft and equipment manufacturers use non-standardized EVSE and connectors (such as the connector in Figure 10). This

Figure 9. Example of MCS Connector



Source: Stäubli

Figure 10. eGSE Connector



Source: Consultant Team

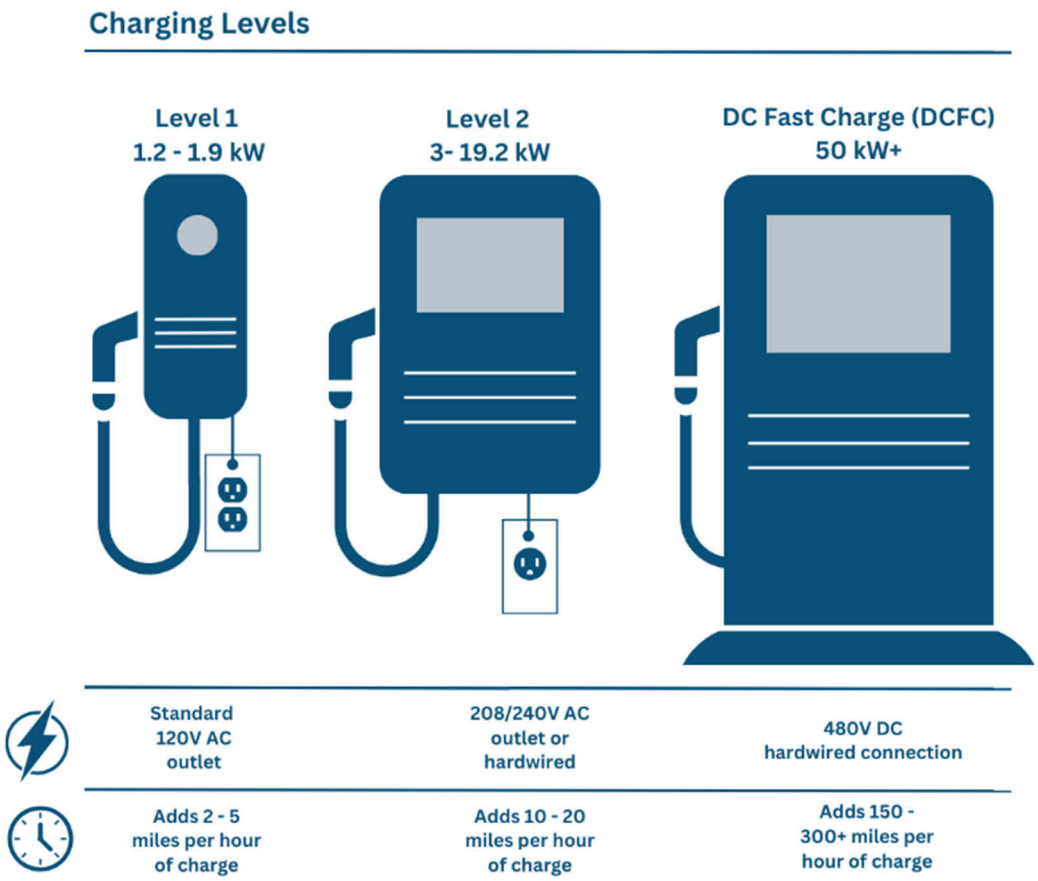
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allows the manufacturer to manage design requirements and charging needs of the specific vehicle. However, non-standard connectors create interoperability challenges, which limit the usefulness of the chargers for a select vehicle, aircraft, or equipment type.[21] The push for interoperability standards for electric vertical takeoff and landing (eVTOL) charging systems is ongoing.[22]

What are the types of EVSE and when is each used?

EVSE is classified by the rate at which they deliver power to recharge an EV battery. There are three primary categories of EVSE, as shown in Figure 11.

Figure 11. EVSE Charging Levels



Source: Consultant Team

Below is a summary of the three primary types of EVSE, charger characteristics, considerations for EVSE selection, and likely use cases at airports.

Level 1 Charging

L1 charging, which uses a standard 120V alternating current (AC) electrical outlet, is the slowest and lowest-cost type of charging available. An L1 charger will provide approximately three to six miles of range per hour of charge for an LDV. Because of its slow charging speed, L1 is typically limited to LDVs that have long dwell times (i.e., the amount of time an EV is parked at a charger). The majority of L1

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charging takes place in residential applications for overnight charging of vehicles that travel less than 40 miles per day.

For L1 charging, typically only a power outlet is provided at the location, with the user providing the EVSE charging cable that connects to the vehicle (Figure 12). L1 charging presents the lowest-cost option for airports and is suitable for long-term parking areas where a vehicle may be left for a day or more. This option may also be suitable for charging employee EVs or airport fleet vehicles with very low daily use. L1 charging is often provided without a user fee. It can also be installed to operate through a charging network, allowing the site host to establish a user fee to collect revenue, though this option is infrequently employed. A summary of L1 EVSE attributes can be found in Table 1.

Figure 12. L1 EVSE port with J1772 Connector



Source: Consultant Team

Table 1. Level 1 EVSE Attribute Summary

Description	Low-speed vehicle charging using a specified charging cable with standard electrical outlet.
Input Voltage	120 volts (V)
Maximum Output	12–16 amps (A)
Power Delivery	1.2 to 1.9 kilowatts (kW)
Required Time to Fully Recharge LDV	40+ hours
Installation Cost	Low (<$1,000 per charger)
Network/Fee	Typically, non-networked and offered for free to users, but can be networked and capable of charging user fees.[23]
Pros	Low investment cost; can provide dedicated public charging that does not require daily vehicle turnover; can be provided to public as an amenity for free or at low cost with minimal infrastructure upgrades.
Cons	Slow power delivery and long charging times. Unsuitable for most use cases at airports; only relevant for LDV applications with low daily use or specialty equipment like forklifts. If L1 chargers are publicly available without user fees, airport electricity costs will rise as drivers seek free charging.
Decision-making	L1 is suitable as a low-cost option to rapidly make charging infrastructure available for public or employee use. L1 is best for high volume installations in parking facilities where user fees are not expected to be collected, vehicles that have very low utilization, or situations with budget constraints.

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Level 2 Charging

L2 charging provides higher-rate AC charging than L1 charging through 240V (residential) or 208V (commercial) electrical service (Figure 13). Level 2 chargers can use a 208V/240V outlet or can have a dedicated circuit. L2 chargers have a typical power output of 3 kW to 19 kW, which translates to 12 to 30 miles of range per hour of charge for a typical light-duty vehicle. Level 2 is the most used charging type for public charging locations, fleet vehicle charging or employee EV charging.

Figure 13. L2 EVSE port with J1772



Source: Consultant Team

The specific charging speed of an L2 charger is determined by the charger power rating and a vehicle's ability to accept higher charging levels. Most L2 chargers pull 30 to 50 amps of current. These installations often require infrastructure upgrades that may include new electrical panels for small installations and transformer upgrades for larger installations, which can add significant project costs. Installation of L2 charging typically requires additional construction and infrastructure including trenching and laying of conduit, pads for charging pedestals or wall mounts, concrete bollards, and signage and other needs. However, federal, state, and utility incentive programs are often available to offset a portion of these costs.

Utilization is the fraction of time each day that an EVSE port is actively charging vehicles. In general, higher utilization leads to faster financial return for the EVSE. For fleets, some high-use vehicles may require a dedicated L2 charger, particularly for overnight charging. L2 charging installations often provide multiple EVSE ports at a single pedestal and can be installed to reach multiple parking spaces, increasing charger usage.

L2 chargers are typically classified as networked or non-networked. Non-networked chargers are typically selected for private charging with no public access and without charging fees. Networked chargers are more common than non-networked chargers in shared use locations and are typically operated by a third party. Networked stations often allow users to pay for charging sessions using a credit card, mobile device, key fob, or simply by plugging in. Additionally, in some instances network providers may cover some or all the construction and installation costs of public L2 charging infrastructure on airport property. Network providers can then recoup the installation cost through electricity sales and ongoing network fees charged to the site host.

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Table 2. Level 2 EVSE Attribute Summary

Description	Moderate-speed vehicle charging using a 240V/208V AC dedicated circuit using a hardwired connection or a NEMA 14-50 outlet.
Input Voltage	208V or 240V
Maximum Output	24–80 A
Power Delivery	3–19 kW
Required Time to Fully Recharge LDV	4–10 hours
Installation Cost	Moderate ($4,000 to $10,000 per EVSE port for hardware and installation with no electrical system upgrades). Federal, state and utility incentives are often available, or infrastructure costs are absorbed by network providers and recouped by electricity sales and network fees. More information available at the DOE's Alternative Fuels Data Center.[24]
Network/Fee	May be offered for private/fleet charging without setting fees; user fees may be set by site host or third party for public/private networked chargers.
Pros	Suitable for most LDV applications, specialty vehicles, and lower mileage or smaller MDHD vehicles. EVSE ports can be used to serve multiple vehicles on a daily or weekly basis. Networked stations allow airports to collect revenue from electricity sales and manage charging
Cons	Can require electrical infrastructure upgrades and construction/retrofits of existing parking areas, increasing capital costs. Conversion of grid AC power to DC power may require an additional transformer at the project site, and large installations may require significant time and capital resources for construction. Operations and maintenance and network fees (where applicable) generate ongoing costs.
Decision-Making	L2 will likely comprise most charging infrastructure at airports. L2 should be selected to serve fleet vehicles that have long dwell times and/or travel less than 60 miles per day. L2 the best option for publicly accessible charging and offers the airport the opportunity to collect revenue from user fees.

DCFC Charging

DCFC (sometimes called Level 3) is the fastest type of charging. DCFC EVSE convert grid supplied AC power to direct current (DC) power to directly charge a vehicle battery (Figure 14). DCFC chargers supply a minimum of 50 kW of rated power and are commonly classified as 50 kW, 150 kW, and 350 kW. These chargers typically use a dedicated 480V AC circuit and require significant infrastructure upgrades and utility planning; however, new DCFC EVSE integrated with battery systems offer periodic high-speed charging at Level 2 power rates (208V/204V).[25]

Figure 14. DCFC EVSE Port with CCS Connector



Source: Consultant Team

DCFC chargers are suitable for high-use LDVs with short dwell times and for vehicles with large capacity batteries such as buses, MDHD vehicles, and aircraft. However, not every EV will be capable of charging at higher DCFC charging rates, such as 150 kW and

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350 kW. A summary of DCFC EVSE attributes can be found in Table 3. Depending on its power rating (and other factors like outside temperature and battery health), a DCFC charger can fully recharge an average-size passenger vehicle in 20 minutes to one hour. There are also applications of higher power DCFC charging of more than 1MW capable of using 1000V DC power, though limited to date.

Because of the extremely high cost of DCFC ports, they are typically installed to serve many vehicles daily. For public charging at airports, DCFC chargers may be used by ride-hail vehicles, buses, drivers in waiting areas or for rental car customers in need of charging. For private charging, DCFC chargers are likely to be used in MDHD applications including transit buses, and in the future for charging of electric aircraft in greater numbers.

Because of the cost of infrastructure, increased power consumption, and demand charges levied by the electric utility, (see: What are demand charges and why are they important?) DCFC vehicle charging is significantly more expensive than L2. While public L2 charging may cost $.15-.35 per kilowatt-hour (kWh), DCFC charging may be $.40-.60/kWh or more.

Table 3. DCFC EVSE Attribute Summary

Description	High-speed vehicle charging using a 480V AC dedicated circuit using a hardwired connection.
Input Voltage	480V
Maximum Output	100–500 A
Power Delivery	50–350+ kW
Required Time to Fully Recharge LDV	20 min – 1 hour
Installation Cost	High ($50,000 to $200,000+ per EVSE port for hardware and installation). Federal, state, and utility incentives are often available.
Network/Fee	May be offered for private/fleet charging without setting fees; user fees may be set by site host or third party for public/private networked chargers.
Pros	The only suitable option for many use cases that require short dwell times or have high demand for power. EVSE ports can be used to serve multiple vehicles daily and can operate similarly to traditional fueling infrastructure, limiting operational charges for vehicle use. Public stations allow airports to collect revenue from electricity sales.
Cons	The highest cost charging type and may produce significant peak demand charges levied by the utility. Projects require significant electrical infrastructure upgrades including new transformers. Projects will require substantial new construction and retrofits of existing parking areas and may result in loss of parking due to footprint of infrastructure. Operations and maintenance and network fees (where applicable) require ongoing investment.
Decision-Making	DCFC will likely be required for many airport use cases that require fast charging times, such as rental cars and for-hire vehicles, and large or high-use vehicles, such as buses, aircraft, or specialty equipment. Because of their cost, DCFC chargers require high utilization, and installation will be driven by customer/tenant demand. Will require significant utility engagement to ensure available capacity or capacity expansion. Project timelines can be one to two years or more.

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What are public and private chargers?

Access to a given EV charger at an airport depends on the charging **use case**. EVSE will generally operate as one of two types:

- **Public chargers** are accessible to all drivers and vehicles. At airports, this may include passengers, taxis, TNCs, and other vehicles and vehicle operators. These chargers will typically be networked to enable communication via an internet connection, which will allow for electronic payment, facilitate software updates, and provide other capabilities.
- **Private (restricted access) chargers** are not accessible to all drivers. These chargers typically serve a specific vehicle or a group of vehicles and have customer- or employee-restricted access. Private chargers at airports may be installed to serve both landside and airside operations, eGSE, airport fleet vehicles, buses, or future electric aircraft. While most private chargers will be networked to enable tracking of energy consumption, charge management, and software updates, non-networked chargers may be preferrable to reduce investment costs.

What are networked or smart chargers?

EVSE can be networked, with a connection to a central backend system via the internet, or they may be non-networked (not connected to an external system).

- **Networked**, or smart chargers (Figure 15), have communications systems that enable information to be transferred via Wi-Fi or cellular networks. Networked chargers typically require an ongoing monthly, per session, or annual networking fee for the user, the site host, or both. Despite the added costs, networked stations provide several benefits over non-networked stations. Networking allows the station operator to track use by time of day, enabling better management of the station compared with non-networked stations. Networking also allows for quicker identification of faults and defective equipment. Finally, certain networks enable advanced features such as charge management, decreased electricity bills, and power sharing between multiple chargers at a site.
- **Non-networked** chargers (Figure 16) act only to deliver power to a vehicle and do not have communications systems. These chargers do not require software and are a lower cost option for airport operators. Non-networked chargers may be suitable for situations that do not require payment or a specific use case without public access.

All charger types can be networked or non-networked. L1 chargers are typically non-networked to reduce costs. Level 2 chargers are a mix of networked and non-networked chargers, depending on the use case. DCFC chargers are almost always networked because of their complex operation; to allow remote monitoring,

Figure 15. Networked L2 EVSE



Source: ChargePoint

Figure 16. Non-Networked L2 EVSE



Source: Enphase

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control of equipment, and collection of charging data, and to enable payment by users.

The charging protocol **ISO 15118** is the international protocol that enables an EV to communicate with an EV charger and the grid. ISO 15118 is important for the widespread adoption of bidirectional charging, which will enable EVs to participate in demand response programs. ISO 15118 is also central to plug and charge systems, which allow a charger to automatically begin a charging session upon vehicle connection without the need for payment or other processes.

Several considerations determine which charger type should be installed at an airport:

- Does the airport want to set a user fee/collect revenue from the charger?
- Does that airport want to control who can use the charger?
- Does the airport want to run diagnostics and monitor charger health remotely?
- Does the airport want to track vehicle charging and other operational data?
- Does the airport want to implement load management or demand response programs?
- Does the airport want to have the ability to upgrade charging software in the future?

If the answer to any of these questions is yes, it is likely that a networked charger is the appropriate charging type.

What is mobile charging?

Mobile charging is an emerging EV charging process that can serve many of the same functions as traditional charging stations with several additional capabilities. Large mobile chargers can service entire vehicle fleets with reduced fixed infrastructure while smaller chargers can provide critical range for long trips or stranded EVs. Mobile chargers have the added benefit of acting as stored energy sources, offering similar grid resilience benefits as traditional energy storage systems. Both Level 2 and DCFC mobile stations are available for procurement on the market today.

Figure 17. L2 Mobile Charger



Source: FreeWire

Mobile chargers can range significantly in size, but all are designed to be deployed to sites where charging is needed on demand. For example, the Tesla Megapack Mobile Supercharger is a 3.9 MWh battery approximately the size of a shipping container that can provide DCFC charging, while the FreeWire Mobi charger (Figure 17) is 80 kWh and provides Level 2 charging capabilities. Mobile charging is particularly useful for applications that require an EVSE to come to the vehicle location. They can also serve as a short-term solution for EV charging needs during construction of permanent charging infrastructure. Because mobile charging may not require infrastructure upgrades

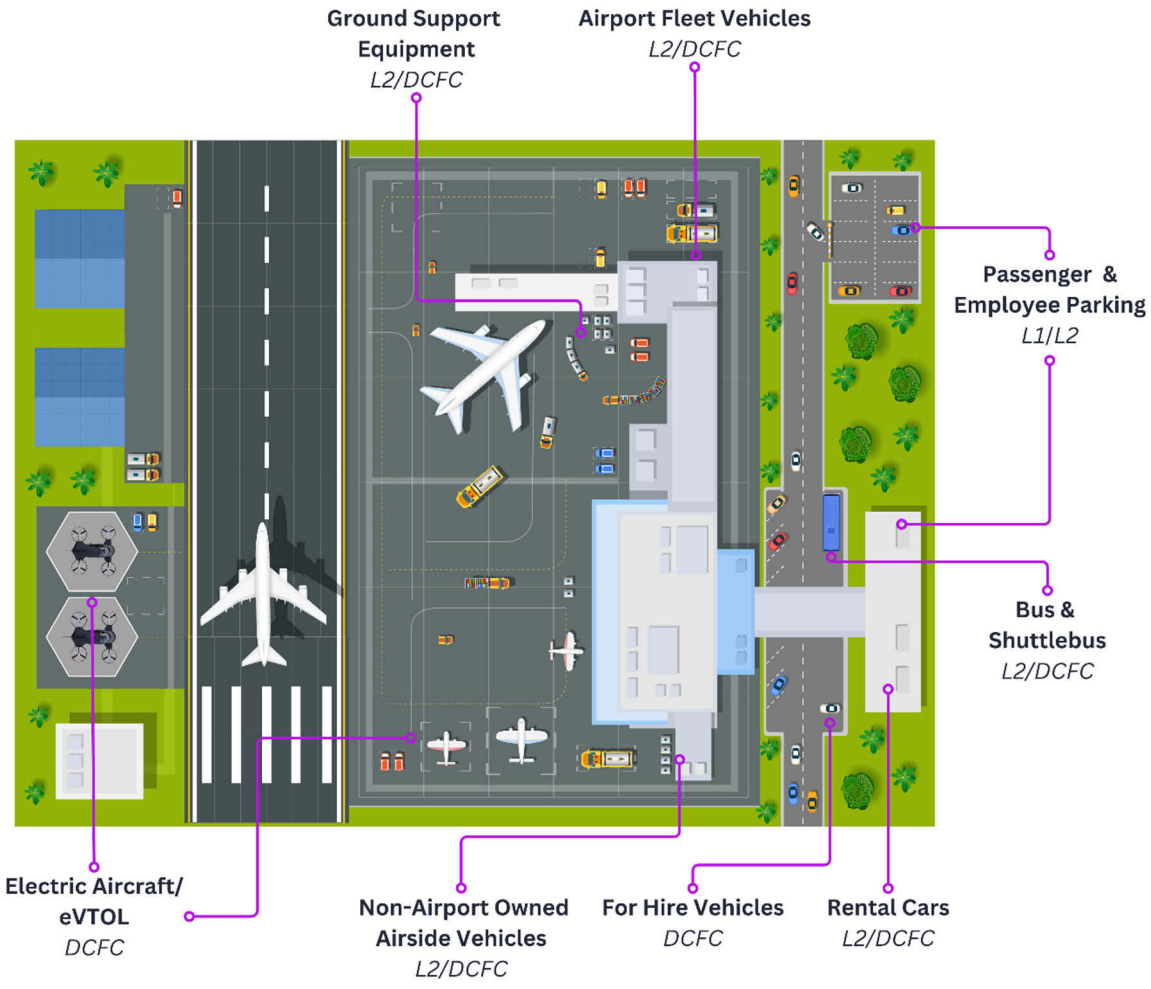
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and does not require trenching or other construct, it can be a cost-effective option compared with typical EVSE installations. However, mobile charging is limited in ability to meet charging requirements at scale.

What are the use cases for charging infrastructure at airports?

In the development of this guide, the consultant team identified eight generic "use cases" of charging infrastructure at airports: Ground Support Equipment, Passenger & Employee Parking, Electric Aircraft, Non-Airport Owned Airside Vehicles, Airport Fleet Vehicles, For Hire Vehicles, Bus & Shuttlebus, and Rental Cars. Each use case represents a distinct vehicle-charger combination with unique operational and charging characteristics. Figure 18 shows where these eight use cases could be located throughout the airport. The following pages summarize key attributes of each use case.

Figure 18. Example Use Cases at Airports



** Please note that this image is for illustrative purposes only. It has not been developed to scale or for planning purposes.*

Source: Consultant Team

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Use Case 1: *Ground Support Equipment*

Description: This use case comprises electrified ground equipment and vehicles that support aircraft and other equipment (often referred to as eGSE). Services include refueling; towing aircraft or luggage/freight carts; loading luggage/freight, potable water, and food; transporting passengers; removing sewage; de-icing airplanes; and fighting fires. Approximately 10% of airport GSE is now electric,[26] with major deployments at many large hub airports such as airports in Boston,[27] San Francisco,[28] Philadelphia,[29] and the Port Authority of New York and New Jersey. An example of eGSE charging at SAN is shown in Figure 19.

Vehicle Types: Tugs, pushbacks, belt loaders, container loaders, luggage tugs, lavatory trucks, and water trucks.

Vehicle Duty Cycle: Most GSE duty cycles consist of short periods of high-load operation followed by periods of no activity.

Vehicle Owner: Airline, third-party operator, or airport operator.

EVSE Owner: Airport operator or tenant.

EVSE Type: Mix of L2 and DCFC.

Figure 19. eGSE Chargers



Source: Consultant Team

Use Case 2: *Passenger and Employee Parking*

Description: This use case includes chargers for passenger vehicles parked at hourly, daily, and long-term lots, as well as airport staff and tenant vehicles in employee lots and will consist almost exclusively of LDVs.

Vehicle Types: Passenger cars and light trucks; possibly motorcycles, mopeds, and electric bikes.

Duty Cycle: An estimated 60% of the U.S. population can reach a commercial hub airport in 30 minutes and 90% of the U.S. population can reach a regional airport within 30 minutes.[30] These statistics suggest that most passengers who park at daily, hourly, or long-term lots do not require a full charge when they reach the airport.

Owner of the Vehicles: Private (passenger, employee, and third party).

Owner of the Charger: Airport operator.

Typical Charging Power: Mix of L1 and L2.

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Use Case 3: *Electric Aircraft*

Description: This use case comprises all electrified aircraft, including fixed-wing aircraft and eVTOLs. Today, only a few electric aircraft models, such as the Pipistrel Velis Electro, which is a two-seater, fixed-wing aircraft used primarily for flight training (see Figure 20), are flying in the United States. Many new electric aircraft technologies associated with the Advanced Air Mobility (AAM) and Urban Air Mobility (UAM) industries are in development, and several are currently undergoing the FAA aircraft certification process.[31] Advanced Air Mobility (AAM) operations in the near-term are expected to primarily use existing airports and heliports.[32] The first electric aircraft in service will likely have small capacities and will target missions for private and freight delivery, recreational flights, training purposes, air taxi services, and regional aviation.[33] Current charging standards for on-road transportation (up to 350 kW) align with multiple light electric aircraft currently applying for certification.[34] A key consideration for integrating electric aircraft into airport systems is ground time needed for recharging without adversely affecting operations.

Figure 20. Examples of Electric Aircraft



Source: ACRP Research Report 236. Militky-Brditschka MB-E1, Pipistrel Velis, and magniX Cessna e-Caravan

Vehicle Types: Conventional takeoff and landing aircraft (flown horizontally), short takeoff and landing aircraft that can use a short runway, and eVTOL aircraft that can operate like helicopters.

Duty Cycle: Aircraft such as eVTOLs are expected to have a unique duty cycle characterized by high discharge currents at the beginning and end of travel (corresponding to the takeoff and landing profiles of aircraft) and a moderate power requirement between.[35] Conventional aircraft are expected to require high battery capacities for local and regional travel.

Vehicle Owner: OEMs, private owners, flight schools, legacy airlines, and startup air taxi/cargo operators.

EVSE Owner: Airport operator, fixed-base operator (FBO), airport tenant (such as a flight school or air cargo operator), airline/flight operator, private owner, and/or specialized third party.

EVSE Type: High power DCFC.

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Use Case 4: *Non-Airport-Owned Airside Vehicles*

Description: This use case includes vehicles that are operated by the airport or by tenants and that are driven exclusively behind the fence. These vehicles typically have very low daily mileage but may have high power demands for materials transport, frequent idling, and specialized service applications.

Vehicle Types: Light-, medium-, heavy-duty, and specialty vehicles.

Duty Cycle: These vehicles are characterized by long periods spent traveling routine paths and idling. Facility and maintenance vehicles have a wide range of duty cycles, from long periods of non-operation (snowplows or mowers) to regular daily use (forklifts).

Owner of the Vehicles: Airport operator or third party.

Owner of the Charger: Airport operator or third party.

Typical Charging Power: Primarily Level 2 chargers with infrequent DCFC for high-use, large vehicles.

Use Case 5: *Airport-Owned Fleet Vehicles*

Description: These are airport-owned and airport-operated vehicles on both the airside and landside (excluding buses and shuttle buses). Services include emergency response and security, facility and maintenance, and airport fleet vehicles.

Vehicle Types: Light-, medium-, heavy-duty, and specialty vehicles.

Duty Cycle: The duty cycles of emergency response and security vehicles are characterized by long frequent periods spent traveling routine pathways and idling, and far fewer periods of intensive driving, sometimes at high speeds. Facility and maintenance vehicles have a wide range of duty cycles, from long periods of non-operation to regular daily use.

Owner of the Vehicles: Airport operator or third party.

Owner of the Charger: Airport operator or third party.

Typical Charging Power: Given the duty cycle, most chargers for fleets are likely Level 2 EVSE with less frequent DCFC EVSE for high-use, large vehicles.

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Use Case 6: *For-Hire Vehicles*

Description: This use case includes chargers for taxis, ride-hail services, TNCs, and other private vehicles transporting airport customers. Airports may become fast charging hubs for taxis and ride-hail services that are looking to extend their daily range and are willing to wait in hold lots between passengers. For drivers waiting to drop off or pick up airport passengers, some airports have deployed DCFC in cell phone lots, providing charging for vehicles on airport-related business and for the public. Beyond airport boundaries, some ride-hail companies rely on charging stations located near airports to support their electric fleet.

Vehicle Types: Light-duty passenger vehicles.

Duty Cycle: The duty cycle for vehicles at airports is characterized by extended periods of waiting for passengers paired with extended periods of highway driving.[36] This pattern suggests that ride share vehicles can benefit from DCFC infrastructure at or near airports to obtain the necessary range to make repeated long-distance trips.

Owner of the Vehicles: Third party.

Owner of the Charger: Airport operator or third party.

Typical Charging Power: The vehicles require a mix of DCFC and Level 2 charging given the unpredictable daily mileage and refueling windows.

Use Case 7: *Bus & Shuttle Bus*

Description: This use case consists of chargers for airport and tenant owned and operated buses and shuttle buses. Buses and shuttle buses should be among the first set of vehicles to electrify at an airport because of the high levels of pollution near pedestrians, the vehicles' predictable duty cycles, and the vehicles' high public visibility.

Vehicle Types: Buses and shuttle buses.

Duty Cycle: Fixed-route operations, varied by vehicle owner and operator.

Owner of the Vehicles: Airport operator or third party.

Owner of the Charger: Airport operator or third party.

Typical Charging Power: Large passenger buses are expected to require DCFC charging. Depending upon duty cycle, some shuttle buses may use Level 2 charging.

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Use Case 8: *Rental Cars*

Description: This use case includes chargers for vehicles rented to travelers by RACs or consolidated rent-a-car (ConRAC) operators. RACs including the Hertz group, Enterprise Holdings, and the Avis-Budget Group have announced commitments to transition to a zero emissions fleet. Hertz currently has 50,000 EVs in its fleet and plans to have 340,000 EVs by 2027. Because RAC typically operate as airport tenants, large-scale electrification of this use case poses significant challenges for airport operations. Electrification of this segment may require increased customer facility charges to be imposed on rental car companies to satisfy ongoing O&M expenses.

Vehicle Types: Light-duty passenger vehicles.

Duty Cycle: Customer use.

Owner of the Vehicles: Third party.

Owner of the Charger: Airport operator or third party.

Typical Charging Power: Given the high turnover of rental cars, it is likely that rental car agencies will require a mix of DCFC and Level 2 charging.

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Chapter 2. Advanced Topics on EV Charging

This chapter addresses questions on advanced topics on EV charging, building on fundamental questions addressed in Chapter 1.

What are best practices for siting EV charging locations at airports?

Across the eight use cases, airports should ask the following questions when selecting a site for a charging location:[37]

- What is the distance to utility-owned and airport-owned electricity infrastructure?
- Will EVSE installation disrupt existing infrastructure (e.g., will trenching be needed)?
- What is the distance to networking and communication infrastructure?
- Is there adequate space for multiple types of users to access and interface with EVSE?
- Can an EVSE plug reach multiple parking spots (e.g., if a non-EV is parked in one spot, can the charger still be used)?
- Is there adequate space to maintain and repair EVSE?
- Is the location prone to flooding?
- Is there available space for installing additional EVSE in the future?
- Should the location provide preference to EV drivers (i.e., closer proximity to the terminal)?

Every location has pros and cons. To reduce the complexity of construction required to install charging stations, airports should strive to balance the questions above. For example, a potential charging location with ample space for EVSE may not be a good choice if extensive trenching would be required to provide adequate power to the site.

What are specific considerations for siting airside EVSE?

Beyond general siting information above, key considerations in the siting of airside charging stations require coordination with FAA airside safety requirements and include the following:

- Connectivity between the landing area and charging stations.
- Availability of facilities and services for operators and cargo.
- Environmental impacts.
- Airspace surface compatibility for the charger and aircraft.
- Safety Area dimensions to define aircraft parking and impacts to existing layout.

Some electric aircraft have wingspans of 50 feet or more. Setbacks and object-free areas must be checked, and aircraft will need room to park after charging. The airport should utilize existing apron space without negatively impacting apron parking. Siting chargers near a fixed based operator (FBO) or general aviation (GA) terminal can provide many of the services aircraft operators may need.

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Key considerations for eGSE equipment, non-airport-owned airside vehicles, and airport fleet vehicles should include the following items for site selection.

- Ground Power Units (GPUs) near the gate or terminal and power supply at the gate (dependent on power availability and age of the terminal) for eGSE.
- Clear pathways of the Airport Movement Area.
- Fire code restrictions of proximity to the buildings.

Operators need to comply with applicable local and federal fire codes with charging infrastructure near the terminal area and tenant buildings. More information regarding codes can be found in Chapter 5.

How should facility managers select the number and type of chargers?

Decisions about the number and type of chargers at a site are driven by several factors. A summary of key considerations that contribute to charger type selection is provided in Table 4.

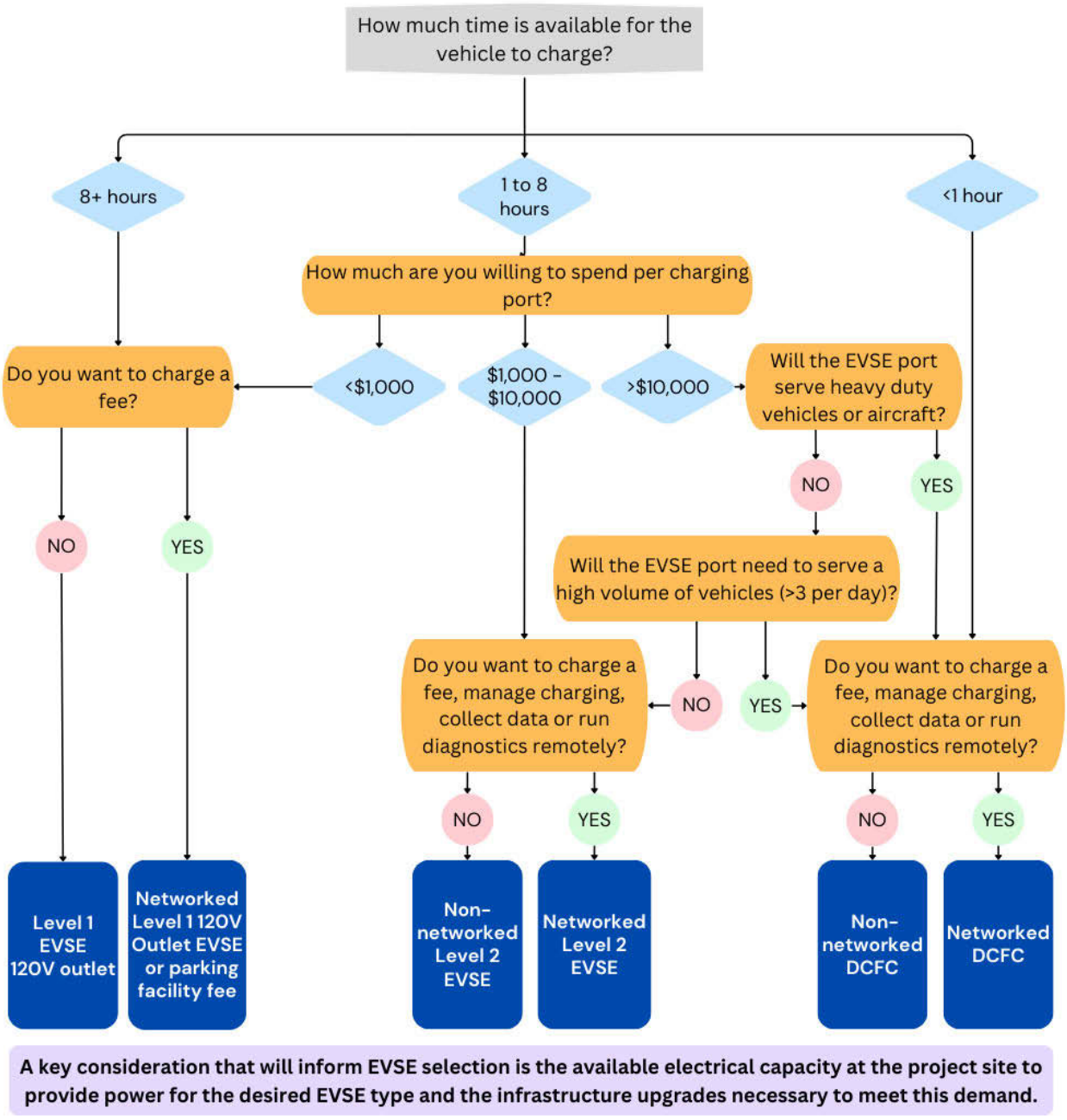
Table 4. Considerations for EVSE Port Selection

Planning Factor	Consideration
How long will vehicles be parked at a charging location (dwell time)?	Longer dwell times can allow lower power charging.
How much funding is available for charging?	Hardware and construction costs increase with higher power charger installation.
What type(s) of vehicles will use the EVSE?	Low power charging is often suitable for LDVs, while MDHDs typically requires higher power charging.
Will revenue be collected from charger users?	Only networked chargers can accept user payment.
How many vehicles will each charger serve daily?	The more vehicles that will be served daily, the higher the power rating of the charger.
How much power needs to be delivered to the vehicle per charging session?	DCFC chargers are required for high power charging.
How much power is available to the charger at the project site?	High power charging may require significant infrastructure upgrades to expand available electrical capacity.
How much physical space is available at the project site for the EVSE, switchgear, transformer, and other infrastructure?	High power charging installations may require additional infrastructure that may reduce available parking or require site redesign.

Approaches to charger selection will vary. Figure 21 provides an example of a simplified decision tree to inform charger selection. Note that for any charging infrastructure project, it is critical to understand the available electrical capacity at the proposed project site and to assess what electrical infrastructure upgrades may be required to support the proposed EVSE.

Figure 21. Example EVSE Decision Tree

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Source: Consultant Team

Who provides EVSE?

There are numerous charging providers operating in the United States that provide various combinations of hardware, network subscriptions, installation, and maintenance. Each company's business model depends on the requirements of the specific site host. For example, the most prevalent non-Tesla EVSP is ChargePoint which primarily provides charging equipment and networking services, while individual site hosts are responsible for installation and maintenance. Other EVSPs, such as Electrify America, provide a turnkey solution to site hosts, whereby the hosts can fully operate and

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maintain the charging infrastructure and set retail electricity prices. Table 5 presents a summary of the 10 largest EVSPs operating in the United States, all of which serve passenger vehicles.

Table 5. Summary of Largest Electric Vehicle Service Providers

EVSP	Level 2 EVSE Ports	DCFC EVSE Ports	Total EVSE Ports	Provides Hardware	Installs and Maintains Hardware	Sets Station User Fee	Provides Cloud Network
ChargePoint	62,669	2,450	65,119	✓	-	-	✓
Tesla	15,019	24,517	39,536	✓	✓	✓	✓
Blink Charging	16,535	151	16,686	✓	✓	-	✓
FLO	6,886	576	7,462	✓	-	-	✓
Electric Circuit	4,070	874	4,944	✓	✓	✓	✓
Electrify America	220	3,835	4,055	✓	✓	✓	✓
EV Connect	2,862	768	3,630	✓	✓	✓	✓
Volta[a]	3,029	115	3,144	✓	✓	✓	✓
Shell Recharge	2,455	620	3,075	✓	✓	✓	✓
EVgo Network	341	2,713	3,054	✓	-	-	✓

[a] Volta is now part of Shell Recharge.

Source: DOE AFDC and Consultant Team, Accessed November 14, 2023

How can airports "futureproof" charging infrastructure?

Futureproofing means building charging infrastructure to accommodate future changes in charging demand or technology. Effective futureproofing reduces costs and saves time. For example, futureproofing includes placing extra conduit and larger conductors to limit the need for trenching if the installed capacity needs to expand or coordinating TTM infrastructure with the local utility to oversize transformers and switch gear. "EV-ready" is synonymous with futureproofing. Specific requirements for EV-ready infrastructure vary by jurisdiction but typically include a combination of increased panel capacity, installation of electrical stub-outs (Figure 22), additional conduit and conductors, and pouring extra concrete pads.

Figure 22. Examples of EVSE Stub-Outs



Source: Consultant Team

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Who owns and operates charging infrastructure?

There are several potential ownership models for charging infrastructure:

- **Site Host Owner-Operator.** In this model, the owner of the property providing the charging infrastructure (such as the airport) owns, maintains, and sets user fees at the charging stations. This model provides the site host control of infrastructure and allows hosts to keep all revenue generated from fees for the electricity sold. This model also increases risk for the site host, who is responsible for ongoing maintenance, technology upgrades, and potential financial losses due to low charger utilization or lack of charge management (see: *What are demand charges and why are they important?*).
- **Utility Owner-Operator.** A site host may choose to partner with the local electric utility for the design, construction, and operation of charging infrastructure. The utility may lease the chargers to the site host or may develop its own sites and charging network. By partnering with the utility, a site host may mitigate the risks associated with maintenance and operation and may also reduce design and permitting delays because of close coordination with utilities. However, the revenue generation potential for the site host is lower than with other models.
- **Third-Party Owner-Operator.** A site host may contract with a third party, such as an EVSP, to handle some or all EVSE ownership, operation, maintenance, and billing responsibilities for the charging stations. An example of this model is the Electrify America charging plaza at the Sacramento International Airport.[38] In this model, the EVSP provides a turnkey solution, and the airport provides the land. There is flexibility in this approach as the two parties are free to negotiate a mutually agreeable set of terms, roles, and responsibilities. With this approach the site host and third-party vendor may enter into a revenue sharing agreement, whereby the host collects a portion of the revenue generated from electricity sales.
- **Charging-as-a-Service (CaaS).** With this business model, a third party provides all capital associated with developing the charging infrastructure. The third party owns the equipment and effectively leases it to a site host under a service agreement, which may also include operations and maintenance. This approach can be beneficial for entities that wish to reduce the upfront capital cost of developing the charging infrastructure by converting this cost to an operating cost that is typically charged as a monthly fee along with additional service charges by the vendor. This arrangement may increase the long-term costs for site hosts because larger fees are spread out over the lifetime of the operating agreement; it may also result in higher fees for charging fleet vehicles that the site host may operate.

What maintenance do charging stations require?

Responsibility for the maintenance of charging stations at airports varies depending on the ownership structure of charging equipment and the specific relationship between an airport, the relevant EVSE installer, and any third-party organizations involved in ownership and operation of EVSE. It is important for airports to establish who is responsible for the operations and maintenance (O&M) of their charging stations and to include details in maintenance contracts regarding response time, repair time, and overall uptime requirements.[39] The U.S. DOE Alternative Fuels Data Center recommends that station

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owners estimate an average maintenance cost of $400 annually, per charger, and that most charging networks offer maintenance plans for an additional fee.

There are several preventative maintenance measures for EV charging stations: [40]

- Install raised concrete pads to limit equipment exposure.
- Place screens away from direct sunlight and use protectors to prevent damage and overheating.
- Install collision protection and signage to prevent and limit damage from accidents.
- Use short cords or cord controls to protect charging cords from people and vehicles.
- Store cords on EVSE dispensers or use cord retractors to increase safety and prevent damage when charging equipment is not in use.

There are also several best practices for the regular maintenance of EV charging stations:

- Shut off power to equipment before conducting maintenance.
- Clean connectors and pins with light detergent and non-abrasive cloth.
- Inspect power to EV charging equipment before conducting maintenance.
- Conduct routine inspections of cord and cord storage devices for wear.
- Clean air conditioning or cooling filters.
- Inspect the mounting area of EVSE for potential compromises like cracks and flooding.
- Review data reports for errors or signs of error in the cloud communication system.
- Compare EVSE data with utility bills to confirm that equipment is functioning properly.

There is no existing standard practice for regular maintenance of EV charging stations. Software, data, and communications systems should be monitored daily to ensure functionality. Inspection of hardware can occur on a weekly or monthly basis.

How are charging fees set?

When the EVSE owner is a different entity than the EVSE user, the owner must determine if and how to assess a fee for usage. There are three general categories for how an EVSE owner sets user fees:

- **Loss Leader (no fee or small fee).** Charging is offered as an amenity for free or for a small fee. At airports, free charging is most often associated with non-networked Level 1 or Level 2 chargers. Typically, this method is used when the EVSE owner wants to increase customer, employee, or tenant satisfaction.
- **Cost Recovery (nominal fee).** The charging fee is set high enough for the owner to recoup installation and operational costs but not to create new revenue. Fees are typically set on a per kWh basis to achieve a lifetime, net present value of zero.
- **Profit Center.** The charging fee is designed to turn a profit from the sale of electricity. Due to demand charges set by electric utilities, it is very difficult for any DCFC charging station to generate enough revenue to cover investment costs on its own without high utilization. Fees are typically set as a price per kWh, per unit of time, or per charging session.

Common pricing structures include charging fees by kWh, by session, by length of time, or through a subscription. In public charging locations, EVSPs offer different pricing models, including member versus

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non-member fees, user-specific pricing (specific to certain vehicle owners), site host–specific pricing, and pricing based on rate of charge.[41] In a standard pay-as-you-go model, drivers pay a set price per minute or per kWh that depends on the level of charging or the amount of energy consumed in a specific charging session. Often EVSPs will collect a nominal fee to initiate a charging session. Most EVSPs offer a membership model, which provides a reduced cost per kWh or per minute for charging and replace session fees with a flat monthly fee. Some EVSPs will also add a fee for leaving a car parked after a charging session has been completed to discourage vehicle idling at charging stations.

Session- and time-based structures are common in states where non-utilities are prohibited from reselling electricity. Over 30 states and Washington DC[42] currently allow per kW sales of electricity for public charging. In some markets, such as California, consumers can take advantage of time-of-use pricing, with which the retail cost of electricity sold is lower during off-peak hours, (e.g., overnight). In states where session- and time-based structures are required, if a charger is dispensing electricity at a slower rate than expected because of faulty equipment, weather, or grid constraints, the actual cost per kWh may be higher.

The cost for charging a vehicle varies for the different charging types. For example, a non-member driver using an Electrify America DCFC station in Texas will pay $0.19 per minute for peak charging speeds of up to 90 kW but will pay $0.37 per minute for a charging speed of up to 350 kW.[43] This pricing reflects demand charges levied by the utility and the increased infrastructure and technology costs for high power chargers.

What is dwell time?

Dwell time refers to the length of time a vehicle is parked at a location. Dwell time is a critical component of the EVSE selection process; the slower the charging speed, the longer the required dwell time. Dwell times can range from minutes to hours to days depending on the charging use case. For use cases such as long-term passenger parking where vehicles may be left for multiple days, L1 charging is suitable and provides a low-cost solution. Conversely, installing L2 chargers in long-term parking areas may decrease revenue for the airport, as vehicles will have long dwell times, which will prevent other drivers from using a charger. In general, public chargers should have the highest utilization possible to increase revenue generation, while private chargers can be operated to match the duty cycle of the vehicle(s) they serve. As highlighted in Chapter 2, it is expected that many use cases of EV charging at airports will have short dwell times (i.e., under one hour), and therefore will require DCFC charging.

What is load shape and why does it matter?

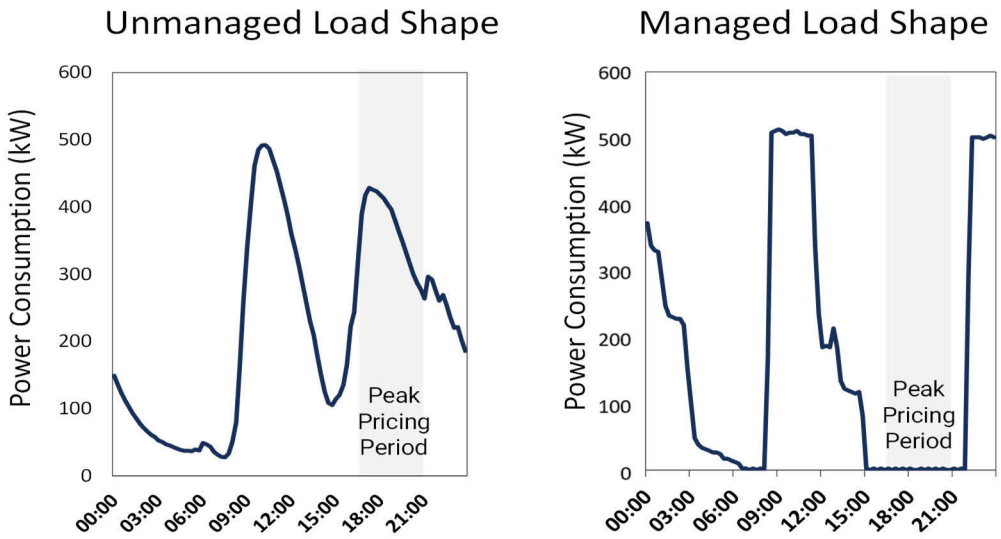
Load shape refers to the profile of electricity consumption resulting from vehicle charging over a specific period, often measured in days, months, or years. Load shape is an important metric for airports and utilities to use to understand demand, forecast the need for infrastructure upgrades, and determine how to implement demand management strategies to smooth electricity consumption and reduce peak load. Concurrent vehicle charging, particularly for DCFC EVSE, can cause large demand spikes, affecting grid stability and increasing electricity costs; these effects can be avoided through charge management.

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Charge management, or smart charging, refers to strategies that can control and optimize the load shape from vehicle charging to minimize grid impacts and reduce costs. Most networked chargers can be programmed to charge at different times of the day or can respond to pricing signals to take advantage of off-peak rates or participate in demand response programs. Many electric utilities offer time-of-use (TOU) rates for EV charging, which incentivizes EV owners to charge vehicles during off-peak periods, such as overnight, which can significantly reduce costs associated with charging.

Figure 23 (left) presents an unmanaged daily load shape of a fleet of vehicles, characterized by peaks in the morning and the end of the workday. Figure 23 (right) shows the same load shape when the fleet manages their charging around the peak pricing period of 4 p.m. to 9 p.m.

Figure 23. Load Shape of Vehicle with (Left) and without (Right) Managed Charging



Source: Consultant Team

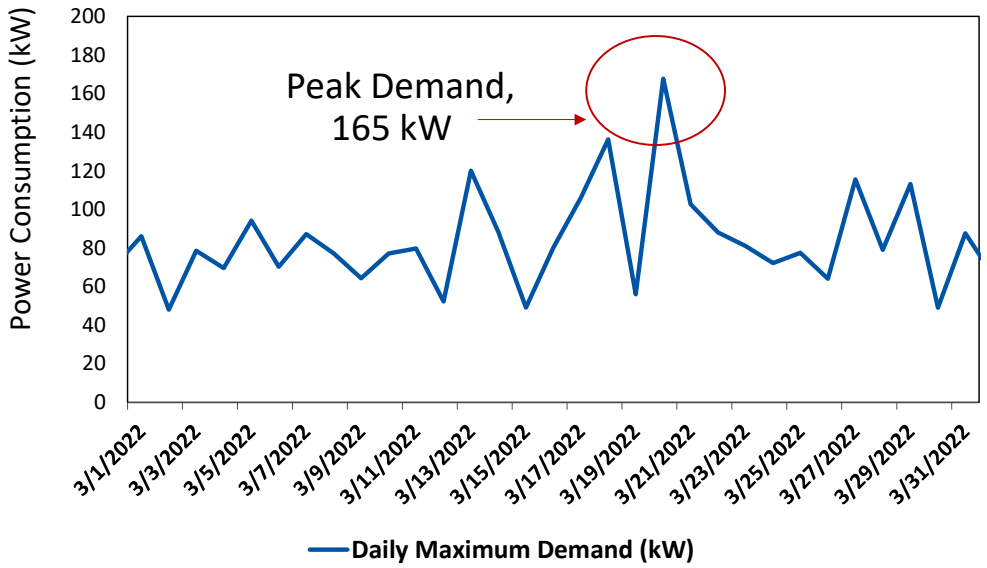
Charge management is not applicable to use cases that have no ability to change when or where they plug in and charge (e.g., emergency response vehicles). While charge management can be employed to reduce costs, it does not prevent the charger from being operated outside of selected time windows. Users can override this programming to charge an EV when needed.

What are demand charges and why are they important?

Demand charges are fees that electric utilities impose based on the peak electricity consumption of a customer over a specific period in a billing cycle. While standard electricity rates and charges are based upon the number of kWh consumed (i.e., $.17/kWh), demand charges are based upon the highest (peak) demand for a given period, measured in kW as shown in Figure 24. Because demand charges are based on peak use within a specific window of time rather than on total energy consumption, the demand surcharge is applied across the billing period. Peak demand may only occur infrequently for a given customer, but the utility must maintain the capacity to meet that demand during any period.

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Figure 24. Example of Peak Demand for Month of March 2022



Source: Consultant Team

Demand charges reflect the need for electricity utilities to build out grid infrastructure and supply electricity to meet peak energy demand, which requires significant additional investment. They help to ensure that costs are distributed to those customers who more significantly impact growth in peak demand across the system. The higher the peak load, the more costly it is for the utility to provide electricity to its customers. Demand charges are also used to encourage customers to shift consumption to periods of lower overall demand, which makes it much easier (and less costly) for grid operators to manage consistent load across time periods and avoid major demand spikes.

Calculating Demand Charges

The basic formula for calculating demand charges is:

$$Demand\ Charge = Peak\ Demand\ (kW) \times Demand\ Rate\ (\$/kW)$$

where:

Peak Demand (kW): The highest recorded energy demand during the specified period

Demand Rate ($/kW): A rate the utility sets that is specific to each customer or customer category, that can vary widely based on the utility's pricing structure, and that may also have time-of-day and seasonal variations.

Electric utilities use different approaches to calculate demand charges, but in general these fees are applied as incremental costs for increased demand. They may be applied per kW ($/kW), by blocks of kW (e.g., $ per additional 10 kW), or using a tiered system (e.g., one rate when demand is 0-1000 kW, another rate when demand is 1,000-5,000 kW, etc.).

Demand charges are particularly relevant for DCFC charging. Unlike L1 and L2 charging, which may have minimal impact on peak demand, DCFC charging requires short periods of high electricity consumption.

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Demand charges can account for 50% or more of total DCFC EVSE operating costs.[44] Managing the use of DCFC EVSE can have major impacts, as concurrent charging of multiple vehicles can significantly impact peak demand.

Example of Demand Charge Calculation

The following is an example of how demand charges might be applied in two scenarios with eight networked 150 kW DCFC EVSE ports installed at an airport ConRAC to charge the fleet:

- **Scenario 1:** EVSE is available to concurrently charge eight vehicles, which occurs approximately three times per month. The peak demand is 150 kW x 8 = **1,200 kW**
- **Scenario 2**: Using managed charging, EVSE is programmed to limit concurrent charging to only two EVSE at a time. The peak demand would be 150 kW x 2 = **300 kW.**

If the utility demand charge is $10/kW, this would mean that this EVSE site would cost **$9,000** more per month to operate in the first scenario than in the second.

What is demand response for EV charging?

Demand response refers to changes in electricity consumption by customers in response to price or reliability signals distributed by the utility. The goal of demand response programs is to decrease grid stress during periods of peak demand and minimize electricity costs for ratepayers. Demand response programs include:

- **Price-based programs:** Changing electricity prices based on real-time supply and demand conditions. Consumers are encouraged to shift demand from high-price low-price periods.
- **Incentive-based programs**: Consumers receive financial incentives for reducing their electricity usage during peak times. These incentives include bill credits, cash payments, or discounts.
- **Time-of-use (TOU) programs**: Electricity rates vary depending on the time of day. Consumers pay more during peak hours and less during off-peak hours, encouraging them to use electricity when it is cheaper (see *What is load shape and why does it matter?*)

Demand response programs are often used to reduce costs associated with EV charging; however, these programs are not available in every state, and they vary by electric utility. In the future, EVs will likely be able to participate in **Automated Demand Response (ADR)** using the open charge point protocol (OCPP), whereby EVSE can be programmed to respond to real-time pricing signals by minimizing consumption during periods of peak electricity demand or increasing charging when the supply of renewable energy on the grid is high without the need for human intervention.

What is bidirectional charging?

Bidirectional charging refers to the ability of an EV battery to both receive and send electricity back onto the grid or to other devices. The most common term associated with bidirectional charging is **vehicle-to-grid** (V2G), but also includes vehicle-to-building (V2B), vehicle-to-load (V2L), vehicle-to-vehicle (V2V) and vehicle-to-home (V2H); inclusively referred to as vehicle-to-everything (V2X).

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Bidirectional charging, or two-way flow of electricity, is a potential benefit for EV operation in the future, particularly for large fleets. Using the proposed ISO 18118 V2G communications protocol, vehicle batteries could be used to send electricity back onto the grid during demand response events (times when signals are sent to customers to reduce load). EV batteries could be charged when electricity prices are low, and that energy could be sold back to the grid when prices are high. If implemented across a large fleet, bidirectional charging could provide a significant resource to shift load and enhance grid stability or could be used to store excess renewable generation in EV batteries and distribute electricity back onto the grid during periods of higher demand. EV batteries could also be used to enhance resiliency by serving as a source of backup power during grid failures or emergency events.

Bidirectional charging and V2G technologies are in the early stages of deployment, and only a small number of EVs have bidirectional charging capabilities, such as the Ford Lightning and the Nissan Leaf. Most EVSE does not have V2G capability today, but this capability is expected to increase in the future.

What is interoperability for EVSE?

Interoperability refers to the ability for EVs and EVSE from different manufacturers and network providers to operate compatibly. This includes the compatibility of hardware, such as EVSE connectors and payment systems, and of communications systems to integrate multiple EVSE types into a central management system (CMS) to monitor and control EVSE. As EV and EV charging technologies continue to evolve, there is a significant push to develop charging standards that can be adopted nationally.

Hardware interoperability for EVSE charging hardware is focused on the customer interface. This includes the common EVSE connectors—J1772, CCS, and NACS. It also includes customer accessibility, ensuring that public EV chargers do not require membership for use or include multiple payment options. Hardware interoperability aims to make EV charging a convenient, consistent process for users regardless of the make and model of vehicle and across EVSE of all types. Interoperability also increases market competition and reduces infrastructure costs by building a unified charging network.

Software interoperability is focused on communications systems. The OCPP is a standard communication protocol used for EV charging. OCPP is a scalable, open-source, vendor-neutral protocol that governs how EVSE communicates with central network servers and is critical to controlling both charging and processing of payments. OCPP enables interoperability between various EV charging stations and CMS to ensure that EVSE manufacturers and charging network operators can create compatible systems that work together. OCPP has not yet been adopted as a national standard, and many EV charging networks currently operate on proprietary protocols, leading to interoperability issues.

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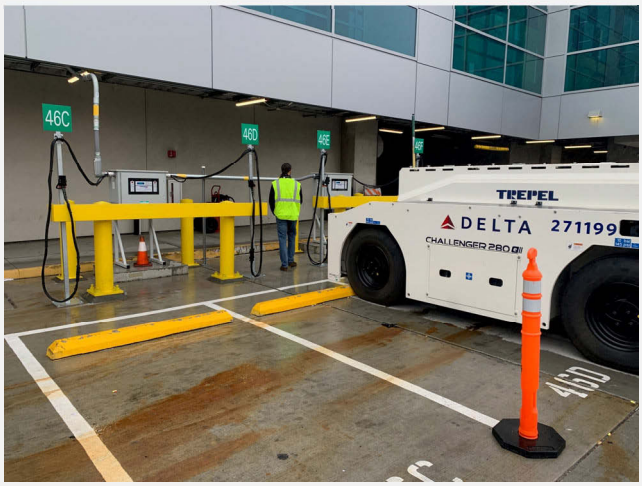
Case Study: San Diego International Airport (SAN)

SAN is a single-runway commercial service airport serving roughly 15 to 25 million passengers per year. In recent years, the airport has moved quickly to advance climate goals, establishing new initiatives like the Good Traveler carbon offset program and achieving Level 4+ Transition in the Airport Carbon Accreditation program—making SAN one of only three North American airports to reach this level. In 2019 and 2020 SAN completed a *Strategic Energy Plan* and a *Clean Transportation Plan* to understand energy consumption and facilitate the transition to vehicle electrification.

The airport began installing charging systems in the 1980s, focusing initially on charging for eGSE. Since then, the airport authority has continued installing eGSE charging at Terminals 1 and 2 (Figure 25). In 2022, SAN installed its latest eGSE using a combination of funding from a Federal Aviation Administration Voluntary Airport Low Emissions (VALE) grant, which covered the purchase of 39 EVSE ports, and incentives from the SDG&E's Power Your Drive for Fleets program, which covered 100% of TTM infrastructure upgrades and up to 80% of behind-the-meter infrastructure upgrades. The airport owns and operates the charging stations on behalf of tenants and wraps utility bills into tenant lease payments using sub-meters. As a general rule, the airport attempts to maintain a ratio of between four and five eGSE EVSE ports per gate.

Airport Tip: Plan charging infrastructure for future demand rather than current needs. Futureproofing, or oversizing charging infrastructure such as transformers and wiring to accommodate additional charging stations, can reduce the cost and labor for future installations when more EVs will be deployed. SAN is working to have 10% to 20% of all parking be EV-capable to accommodate additional EVs over the coming decade.

Figure 25. eGSE Charging Equipment at SAN



Source: Consultant Team

Planning for landside charging stations at SAN depends on the use case. As with the eGSE chargers, the San Diego County Regional Airport Authority owns, operates, and maintains all landside charging. For passenger EVs parked in passenger lots, the airport aims to make 10% of all parking spaces EV ready, meaning the space is fully wired and ready for a Level 2 charging install. The airport staff plans to observe EV adoption over time and install chargers to slightly outpace demand, up to the 10% limit. Charging for transportation network companies and taxis is not available at the airport due to a lack of distribution capacity for high-powered chargers.

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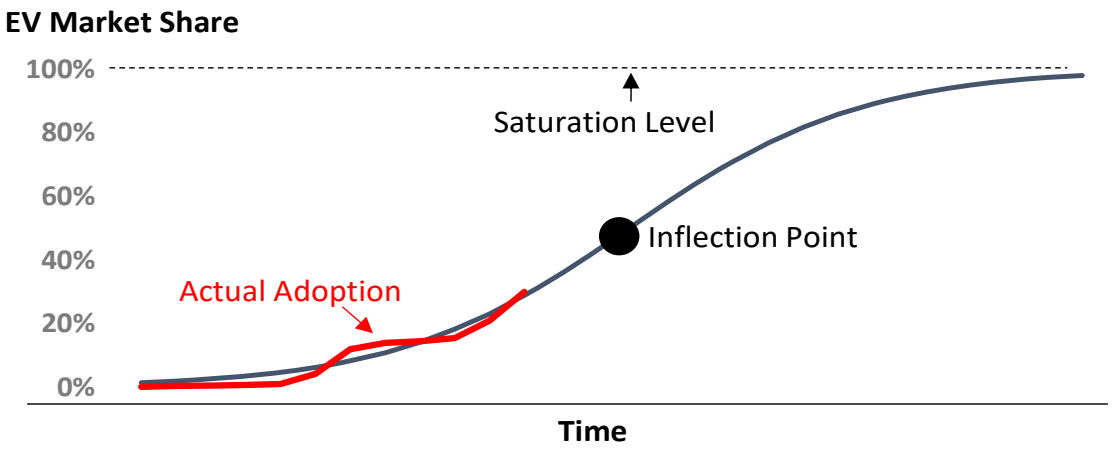
Chapter 3. Estimating EV Adoption, Charging Need, and EV Load

Airport planners need simple methods to predict future EV charging needs and load. Accurate estimation helps right-size infrastructure investments, which leads to efficient use of budget and physical space. This chapter describes methodologies airport planners can use to estimate EV charging needs and load, focusing on light-duty passenger vehicles and employee vehicles.

How can airport planners estimate the future number of passenger and employee EVs?

For airport-owned vehicles, future EV adoption is relatively easy to predict because vehicle procurement is controlled by the airport. For non-airport owned vehicles, such as privately owned LDV, airport staff must estimate the number of future EVs to plan for increased charging needs and load. A comprehensive review of EV prediction methods by Gnann et al. in 2018 describes several potential approaches to approximating EV growth.[45] A common approach uses an "S-curve" function that starts flat, curves upward, then asymptotes at a saturation level (Figure 26). Most new technologies follow this same basic shape.[46] The callout box below provides an example of how to construct an S-curve using a logistic function.

Figure 26. Example of S-Shaped Adoption Curve



Source: Consultant Team

S-curve models are particularly useful if there are already EV sales data available for a region so that early years of the curve can be fit to the data. Simple S-curve models also have drawbacks, including the need to assume (rather than calculate) the peak of sales and the ultimate market potential.

More sophisticated approaches to forecasting EV growth attempt to mimic the decision-making of individuals or groups, while accounting for interactions and feedback. The most common models for EV adoption include discrete choice and agent-based models. However, for airport applications, these models may require too much time and too many resources to efficiently forecast EV growth for multiple airport use cases.

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Example Equation: S-Curve (Logistic Curve)

The equation below results in an S-shaped curve that mimics EV adoption shares.

$$S_i = \frac{L}{(1 + e^{(-k*(x_i - x_0))})}$$

where:

EV Share in Year i (S_i): Percentage of vehicles that are EVs in year *i*. This is the value of adoption.

Maximum EV Share (L): Maximum percentage of vehicles that are EVs (e.g., in 2050).

Steepness of Curve (k): Slope of the adoption curve. High *k* values result in faster ramp up.

Crossover Point (x_0): Calendar year S-curve reaches its inflection point.

Year of Analysis (x_i): Calendar year for estimate (e.g., 2026).

To calculate the EV share in 2026, replace the subscript "i" with 2026. Assume values of 1.0 for L, 0.3 for k, and 2028 for x_0. The equation predicts an EV share of 35.4% in 2028.

How can airport planners estimate the optimal number of chargers?

Typically, the next step after calculating future EV adoption is to estimate the number of EVSE ports needed to support the EVs. These calculations are both an art and science. Most planners use simple ratios such as EVs per charger or percentage of parking that is EV-capable to approximate the right number of chargers.

In general, fleet vehicles use a ratio of one port per vehicle.[47] This ratio minimizes the risk of vehicle downtime because each vehicle is guaranteed a place to charge at all times. Yet, there are some exceptions. Airport personnel interviewed for the preparation of this guide stated that electric GSE ports can be shared and recommended using a ratio of between three and four ports per gate. Transit buses and shuttlebuses often have a mix of high and low power chargers with ratios of EVs to chargers of less than 1.0 because of their relatively high daily mileage.[48]

For passenger and employee vehicle charging, the ratio of vehicles to chargers depends on a variety of factors, including the dwell time of vehicles, the distance to the airport from population centers, and the airport's interest in providing an additional amenity to airport users. The 2021 International Energy Conservation Code (IECC) requires new commercial developments with two or more parking spaces to include at least two EV-ready spaces, and developments with 26 or more parking spaces would be required to also make 20% of all spaces EV-capable, which is defined as having electrical panel capacity and space to support a branch circuit to each parking space. One airport staff member interviewed for this guide said 10% of parking spots in passenger lots were EV-capable, and that the airport was simply installing chargers in those spots as demand materialized. In that way, the airport is right sizing the chargers over time. The City of Boston has perhaps the most aggressive EV charging requirements. In 2019, the City of Boston announced that 25% of parking spaces in new off-street parking areas downtown must be equipped with EV chargers, and the remaining 75% are required to be EV-capable.

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How do regulatory drivers influence EV adoption?

In certain states, EV adoption is largely influenced by the regulatory environment. The key regulatory drivers of EV adoption for airport use cases are shown in Table 6. California has by far the most stringent regulations.

Table 6. Key Regulatory Drivers of EV Adoption, by Use Case

Regulatory Driver of EV Adoption	Ground Support Equipment	Passenger & Employee Parking	Electric Aircraft	Non-Airport-Owned Airside Vehicles	Airport Fleet Vehicles	For-Hire Vehicles	Bus & Shuttlebus	Rental Cars
Advanced Clean Cars II. Requires automakers in certain states to deliver an increasing percentage of cars that are electric. *CA, CO, NY, WA, OR, VT, MA, RI, NJ, MD*		✓			✓	✓		✓
Advanced Clean Trucks. Requires vehicle manufacturers to deliver increasing percentages of MDHDs that are electric within the state. *CA, CO, MD, MA, NM, NJ, NY, OR, VT, WA,*	✓			✓	✓			
Advanced Clean Fleets. Requires fleet operators (businesses and government) to purchase an increasing percentage of MDHDs that are electric. Only in California.	✓			✓	✓			
Innovative Clean Transit. Requires all public transit agencies to transition to a 100% zero-emission bus fleet for new vehicles by 2029. Only in California.							✓	
In-Use Off-Road Diesel-Fueled Fleets Regulation. Requires the phase-out use of the oldest and highest polluting off-road diesel vehicles and encourages the adoption of zero-emission technology. Only in California.	✓							
California Airport Shuttle Regulation. Requires airport shuttle operators to transition to 100% ZEV technologies beginning in 2027 and through 2035.[49] Only in California.							✓	
Clean Miles Standard. Regulation to require electrification of ride-hailing companies starting in 2023 with annual targets requiring zero grams of CO_2 greenhouse gas emissions and 90% of passenger miles traveled to be fully electric by 2030.[50] Only in California.						✓		

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How can airport planners estimate electrical load from EV charging?

Calculating expected load from EVs entails estimating the cumulative load shape across all chargers. A precise load curve is difficult to create because vehicles typically begin and end charging at different times throughout the day in an unscheduled manner. In addition, for a given charger, the power used varies through a charging session (typically highest at the beginning and lowest at the end) and depends on the beginning state-of-charge of the battery. Ideally, rather than creating a hypothetical load curve, airport planners would observe charging load over a given period (e.g., six months) to get a sense of when and where drivers are charging.

An approximate load can be calculated by multiplying the number of vehicles that are charging at a given facility at a specific time by an average power level at which they are charging. This guide does not provide rules of thumb to make this calculation. The callout box provides an example calculation for maximum potential load from EVs.

Example Calculation: Maximum EV Load

A back-of-the-envelope calculation of maximum potential EV load is a useful first step. This calculation is not intended for design or engineering purposes but provides an upper bound that can be refined.

There are several potential methods for calculating EV load, and no method is the best. As an example, suppose there are 5,000 passenger vehicles parked in the long-term parking lot on the airport's busiest day of the year. Also, assume the following:

- By 2040, all light-duty passenger vehicles are electric.
- The number of vehicles parked at the airport will grow by 5% per year between 2025 and 2040.
- 10% of the vehicles need a charge when they arrive at the airport.
- 50% of the vehicles need to charge during the peak period.
- All vehicles use a Level 2 charger with an average power output of 10 kW.

Under these assumptions, the vehicle population would be 5,000. Growing at 5% per year means the population in 2040 is 5,000 * (1.05) ^[15], or **10,400 vehicles**. The anticipated EV load in 2040 would be 10,400 vehicles * 10 kW * 10% * 50%, or **5.2 MW**.

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Chapter 4. Steps to Install Charging Infrastructure

This chapter provides an overview of the basic steps required for the design, construction, and operation of charging infrastructure projects.

What is the process for airports to install charging infrastructure?

Each charging infrastructure project will have unique criteria for development, design, construction, and operation of chargers based on the use case and the partnership approaches for station ownership. However, there is a common set of steps that will be applicable to any charging infrastructure project. These steps are presented below (Figure 27), and a more detailed description of federal, state, and local requirements for charging stations can be found in Chapter 5.

The timeline for construction and activation of a charging infrastructure project can vary widely. For small projects that do not require significant electrical upgrades, the entire design, permitting and construction phase may be completed in six months to a year. Large projects often take at least 18 months, and in many cases may take two years or more. Project developers experience delays due to supply chain issues for transformers, switchgear, panels, and other electrical equipment.

1. Assess Needs and Develop Project Scope

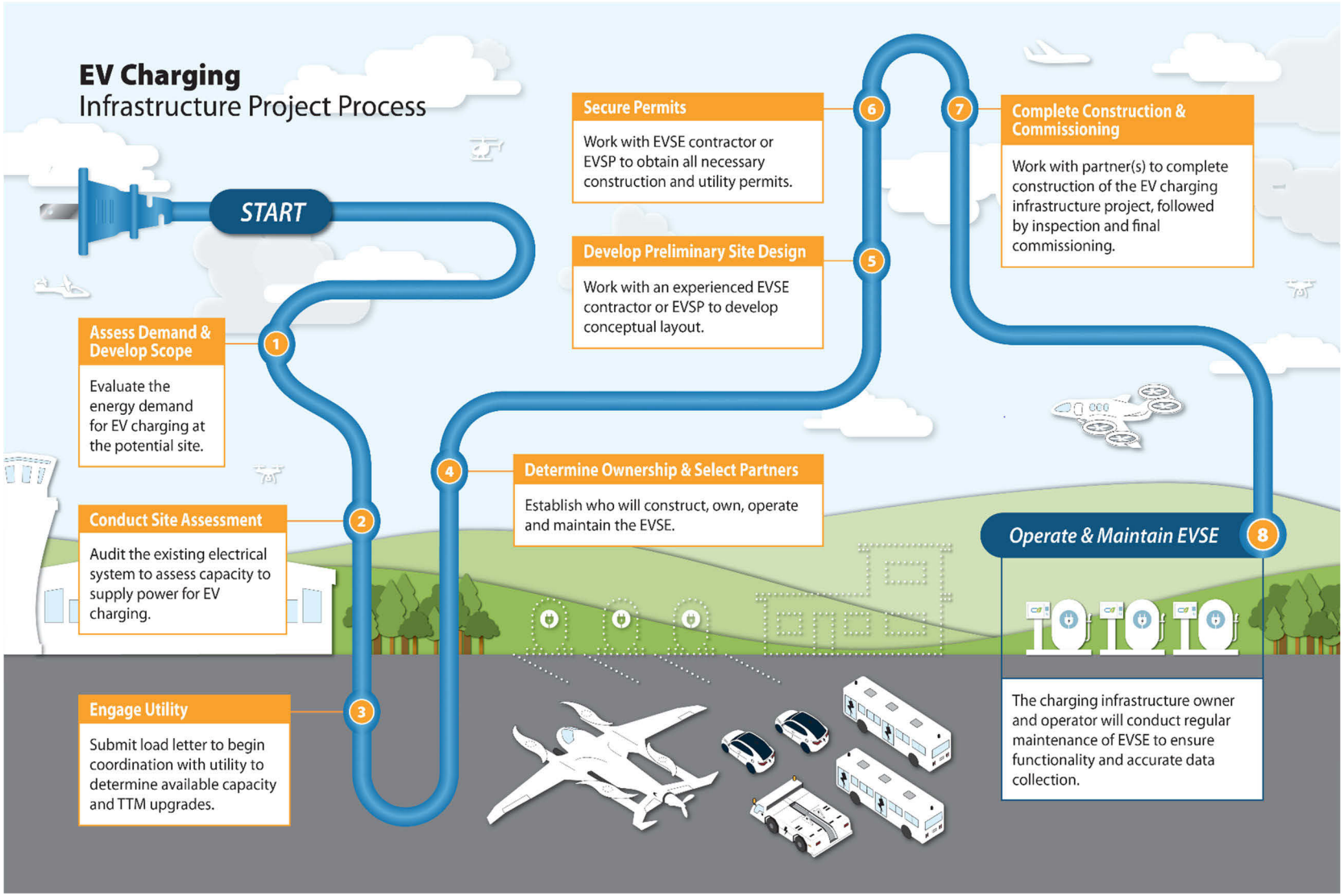
As a first step, the site host needs to understand the demand for EV charging at a potential charging infrastructure site. This will inform both the number and the power level of chargers. The site host must consider charger utilization and energy consumption. Questions to be considered include:

- Where on the property are the chargers expected to be located?
- Is supporting grid infrastructure (e.g., transformer) located nearby?
- What is the capacity of supporting grid infrastructure for additional load?
- Will this be a public or private charging site?
- How many vehicles are expected to use each charger daily?
- What time of day will EVSE be used?
- How quickly will vehicles need to recharge or how long will vehicles be parked?
- How many vehicles will this site serve in the future?
- What is the utility rate structure that the airport is subject to?

The site host should develop a scope for the preliminary project scale, budget and timeline and determine what funding mechanisms will be used to finance the project. The scope should include assessments of any federal, state, local, or utility grant or incentive programs that may be used to offset

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Figure 27. EV Charing Infrastructure Process



Source: Consultant Team

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project costs. The site host should also establish what type of ownership structure(s) could be utilized and whether network capabilities will be required, including payment collection.

2. Conduct Site Assessment

Airport managers should communicate with airport facilities personnel, an EVSE contractor/electrician, and/or an EVSP to conduct an audit of the electrical system to assess the property's existing capacity. This audit will investigate:

- Location(s) of transformers and electrical panels on airport property and distance to the proposed location of the charging infrastructure.
- The output voltage of transformers and electrical panels (120/208V, 120/240V or 480V).
- Approximate current loads on transformers and electrical panels.
- Approximate excess capacity of each existing transformer and electrical panel.
- Approximate space in the area where an existing transformer or electric panel is located. Space may be needed for additional electrical equipment, such as a step-down transformer, subpanel, or switchgear.

The results of the assessment will enable the site host to determine whether the existing electrical system is sufficient to serve potential EV charging demand and understand the costs for upgrades or additions to electrical service that will be necessary to meet increased demand. Planning for EV charging must be coordinated with additional energy audits, load studies, or other projects at the airport to ensure that the available capacity has not already been committed or if additional capacity/infrastructure expansion is already planned for other projects. Energy studies may also reveal opportunities to reduce existing demand, which will increase available capacity for future EVSE installations.

3. Engage Utility

Utilities play an essential part in the rollout of charging infrastructure, and they are among the first partners to consider for electric charging station installations. Working with a local electric utility can be a time-intensive process. Communicating with the utility as early as possible in the EV project planning process is recommended to help inform site design and project timeline. In addition, many utilities offer grant and incentive programs or may identify additional funding sources that may be applicable to a specific charging infrastructure project (see: *Utility Funding*)

The electric utility will need to assess the existing infrastructure's ability to meet the service needs for the proposed EV project. This may require the utility to upgrade infrastructure like transformers, substations, wires, and other equipment. Some of these replacements may involve long lead times that can cause significant delays in project timelines and increase overall project cost. The utility may also be able to provide insight regarding the optimal layout for site design for both TTM and BTM infrastructure, and required electrical capacity upgrades may influence the time and scale of the proposed project.

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A small project with no infrastructure upgrades may only require utility notification, but many charging infrastructure projects will require significant analysis, site inspections, construction and collaboration between the site host and designers, construction contractors, EVSPs, and local government officials.

Airport Tip: Coordinating early with the electric utility on the airport's charging infrastructure program/projects can significantly streamline approvals, timelines, and costs.

Most large hub airports have their own medium-voltage electric distribution system on airport property, acting similarly to a distribution utility. Many airports also already primary power generators, such as through a cogeneration plant, or may have deployed DERs such as solar photovoltaics. Airports have also begun to look closely at the deployment of microgrids—localized groups of interconnected loads and DERs that act as a single entity to enhance resiliency for airport operations. These characteristics will influence the complexity of utility engagement.

While each utility may have its own process, the common steps required for working with the electric utility for planning grid capacity to support airport charging infrastructure includes:

- Based upon the assessment in steps 1 and 2, apply or "load letter" defining the load capacity requested with the electric utility to begin the utility engagement process, determine if utility upgrades will be needed to support the airport's project, and what the timeframe will be. The electric utility will design the TTM infrastructure (everything up to and including the meter) and can also provide advice on incentives and charger rebates.
- Ensure that the charging infrastructure project is coordinated with other electrification programs at the airport. The utility must understand increased demand from the charging infrastructure project in the context of overall airport load growth.
- Work with the electric utility to assess options for futureproofing your project. It is likely that the project you bring to the electric utility will be one of many over time. To efficiently respond to growing demand for charging infrastructure, it may be prudent to plan for larger future upgrade costs at the onset.
- After obtaining permits (Step 6) the electric utility will finalize the TTM design and prepare for infrastructure construction. For some utilities, the utility provider will build the TTM infrastructure after completion of BTM construction; in other cases, the utility may engage in TTM infrastructure upgrades in tandem with BTM construction by the airport. complete and the electric utility will issue qualifying rebates.

4. Determine Ownership Model and Select Partners

Airport operators will need to determine who will own, operate, and maintain EVSE; which will typically be the utility customer. The utility customer may be the site host (airport/property owner), an airport tenant, or a third-party EVSP. As discussed in Chapter 3, there are various models available depending on the specific priorities of the site host, including their ability to set fees and collect revenue, appetite for risk, and desire for involvement in EVSE operations and maintenance.

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Selecting the appropriate partner will have a significant influence on both the cost to the site host and the level of effort necessary for implementation of the charging infrastructure project. EVSP and utility partners can provide insight into EV charging technology selection; support site assessment and design; coordinate with the utility; facilitate grant and incentive applications; and oversee construction, operation, and maintenance of the charging infrastructure as needed.

5. Develop Preliminary Site Design

There are several factors that must be considered when choosing a charger location. The site host should work with an experienced EVSE contractor or EVSP to develop a conceptual layout for the charging infrastructure project. Key items that will need to be assessed include:

- Airport property layout
- Number of parking spaces and chargers to be installed
- Selection of EVSE power rating (Level 1, Level 2, DCFC)
- Distance to electrical service source
- Footprint of additional electrical infrastructure (electrical panel, switchgear, transformer)
- Weather considerations
- Physical barriers to protect the charging stations (e.g., concrete bollards)
- Parking and charging accommodations for people with disabilities
- Lighting and security cameras
- Future expansion capacity for additional chargers
- Incorporation of energy storage, distributed generation, and/or microgrids
- Wi-Fi connectivity for networked chargers
- Equipment requirements to facilitate installation, such as cranes, trenching, and directional boring

Preliminary site design should be inclusive of all components of BTM infrastructure and should provide an understanding of all on-site energy needs. EVSE selection may require an electrical upgrade, including a new meter and electrical panels or switchgear. The size of this equipment will depend on its capacity and installation may require modifications to existing buildings or surface lots. It is recommended that site hosts plan for expansion of EV charging during site planning by oversizing infrastructure and laying additional conduit. This preparation will reduce the difficulty and the cost of adding chargers in the future.

6. Secure Permits

Each jurisdiction will have specific permits that must be secured from state or local agencies. Most EV charger projects are treated as a typical electrical permit and expedited approval may be possible. The airport site host should work with the selected EVSE contractor or other operating partner to obtain all necessary permits, including permissions from the local utility. Ensuring that charger installations are properly designed and installed by experienced contractors can reduce delays in the permitting and inspection process.

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When an airport site host develops project permit and construction timelines, the timelines should include additional time for a possible permit revision and equipment delivery to maintain planning flexibility and minimize financial risk. Once a permit is approved, a final site visit with a building inspector prior to construction and commissioning.

7. Complete Site Construction and Commissioning

Once approvals have been obtained, the site host will work with its partner(s) to construct the charging infrastructure project. The partnership approach for construction of charging infrastructure will be dependent upon the ownership model. As identified in Chapter 3, a site host can work with different vendors for the design, build, operation, and maintenance phases of the project or with a turnkey provider that will be able to guide the site host through each step of the process. Many EVSPs will provide an all-inclusive solution to charging infrastructure planning, including utility coordination and project construction. This will include installation of EVSE charging equipment and construction of electrical infrastructure from the panel to the EVSE. In parallel, the utility will complete TTM infrastructure upgrades required for operation of the EVSE.

Upon completion of the construction phase, EV charging projects will require inspection in accordance with municipal, utility and airport/FAA regulations. Upon approval, the EV project will be activated and commissioned for use.

8. Charger Operations and Maintenance (O&M)

Charging stations require ongoing maintenance to ensure that chargers are operating correctly. This includes physical inspection of cords, connectors, and cord storage; regular cleaning; tracking of network conductivity and charger uptime; and power supply inspection to verify the charging station's voltage, current, and power ratings. Charger repair can be costly if not under warranty, so it is important to determine who will be responsible for the costs and to specify expectations (e.g., response time) in maintenance contracts. Training of employees regarding the operations and maintenance of EVSE may be required for those site hosts who chose to act as owner-operator of EV charging projects.

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Case Study: Augusta Regional Airport (AGS)

AGS at Bush Field is owned and operated by the City of Augusta and is the third largest airport in the state of Georgia.

The City established Skyworx Innovation Xchange (Skyworx) at AGS as an aerospace business incubator designed to help facilitate entrepreneurship and innovative business development in the aerospace industry. In 2018, Georgia Power helped kick off the Skyworx project with a $10,000 Economic Development Improvement Grant.

Airport Tip: When engaging with the utility provider, remember that the airport is a huge user of electricity already, and the utility company is likely willing to support utility expansion at the airport because the airport is a good customer.

In support of the Skyworx program, AGS later installed new electric charging stations on multiple sites throughout the airport through a partnership with Georgia Power and BETA Technologies (BETA), an aerospace manufacturer headquartered in Vermont that develops electric aircraft and chargers. The charging infrastructure project was made possible through Georgia Power's Make Ready Program, which helps customers install affordable electric charging infrastructure so their business can be prepared for the growing electric transportation market and support sustainability efforts. Through this program Georgia Power can provide funding assistance to business customers with emission-free electric transport needs including passenger vehicles, transit buses, school buses, and other technologies like electric aircraft.

BETA installed the AGS chargers in partnership with Georgia Power. The aircraft chargers support fast charging up to 350 kW, and the installations represent BETA's first charging stations in both Georgia and the Southeast. The aircraft charging station includes one DCFC site and a second site that is wired and "future ready" for another DCFC as shown in Figure 28.

Figure 28. AGS Future Ready Site



Source: Consultant Team

At the time of this writing, AGS's charging infrastructure includes:

- Three upgraded transformers to support EV and electric aircraft charging.
- One DCFC site and one future DCFC site for electric aircraft.
- Three EV dual-port Level 2 smart EVSE located in the short-term and general aviation parking lots designed for use by employees and the public.

BETA will own and operate all charging at AGS. All charging infrastructure was installed without any costs to the airport.

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Chapter 5. Navigating Federal, State, and Local Requirements

This chapter summarizes government requirements related to EVSE installation, including regulations, codes, and permitting procedures.

What are FAA planning requirements related to charger installations?

Many public-use airports accept federal grants or federal property transfers for airport purposes. These federally obligated facilities are bound by Federal Aviation Administration (FAA) requirements through its Airport Compliance Program. Federally obligated airports are highly regulated enterprises, and future development of the airport requires FAA coordination, certain FAA approvals, and/or compliance with federal regulations. Several types of airport planning efforts requiring FAA involvement or adherence to FAA regulations include:

- **Airport Master Plan**: FAA Advisory Circular 150/5070-6B, *Airport Master Plans*, provides guidance on the preparation of airport master plans that range in size and function from small general aviation to large commercial service facilities. The goal of a master plan is to provide the framework needed to guide future airport development that will cost-effectively satisfy aviation demand, while considering potential environmental and socioeconomic impacts. While close FAA coordination is maintained throughout the master plan process, the FAA has approval authority over two key elements of a master plan: aircraft activity forecasts and the Airport Layout Plan (ALP). The activity forecast will need to consider the deployment of new electric aircraft in the short term and escalated growth of electric aircraft as the market matures. The ALP is a scaled, graphical presentation of existing and future airport facilities, locations, and relevant clearance and dimensions. The ALP is a primary component of an airport master plan and is used by the FAA to program future funding assistance and to monitor the airport's compliance with design standards and grant assurances.
- **Electrification Master Plan**: Airport operators are increasingly aware of the need to conduct a focused electrification study to guide their transition to electric power systems. The scope of an electrification master plan can vary but typically includes an energy demand analysis, power supply and distribution assessment, infrastructure siting, code compliance, and cost analysis among other components. The electrification master plan must comply with FAA regulations when siting future infrastructure. Coordination with the FAA is advised but not required when these studies are not federally funded.
- **Project-Specific Plan**: Often, infrastructure development is spurred by a particular need of or request from an existing tenant or future business. In these cases, site planning will be driven by the project proponent's needs and preferences but will require significant involvement and oversight by the airport operator to ensure compliance with FAA requirements. Development proposals on the airside area (e.g., the area nearest a parked or maneuvering aircraft) will require the greatest scrutiny and involve the following FAA approvals:
  - **FAA Form 7460-1 Approval**: FAA Form 7460-1, *Notice of Proposed Construction or Alteration,* is required to be submitted to the FAA for any project proposed on airport property. The notice initiates an aeronautical review by the FAA to determine if proposed

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structures may negatively affect navigable airspace. A best practice is to submit this form early in the site planning process to confirm that the proposed site is a viable option for charging infrastructure and not unduly restricted by airspace factors.

- **ALP Pen-and-Ink Change**: Airport operators must maintain an up-to-date ALP, and all proposed projects must be reflected on a FAA-approved ALP before they can be constructed. The ALP pen-and-ink change enables a project-specific update in lieu of undergoing a comprehensive update of the ALP.
- **FAA Section 163 Approval:** When an airport operator submits an ALP change with a request to change a land use from aeronautical to non-aeronautical, the FAA must determine whether the proposal is subject to the agency's approval authority, as defined/limited by Section 163 of the FAA Reauthorization Act of 2018. The airport operator will need to work with the FAA planner to determine if Section 163 would apply to the proposed charging infrastructure project.

What are other Federal requirements for charger installations?

Environmental Review

The National Environmental Policy Act (NEPA) and special purpose environmental laws and regulations require environmental impact analyses of proposed airport actions subject to FAA decision, including federal funding (see: *Federal Funding*). The airport operator will need to communicate with its region's FAA Airport District Office's Environmental Specialist to determine the level of environmental evaluation that is needed, such as a Categorical Exclusion (CATEX), Environment Assessment (EA), or Environmental Impact Study (EIS). The FAA will determine its level of environmental oversight as part of the Section 163 process as described in the FAA Reauthorization Act of 2018 and the NEPA evaluation process. EVSE is generally nonintrusive and not subject to extensive environmental review. However, depending on the scale of the project and required infrastructure upgrades, consideration may be required for extending utility lines to the charging site, particularly for the electrification of aircraft. The Section 163 process requires understanding how the property where the infrastructure will be built was acquired and making this information available.[51]

Flood Prone Areas

An important environmental consideration for charging stations is recognizing and adhering to flood resiliency requirements. Standard industry best practice currently aligns with the Federal Emergency Management Agency's (FEMA) National Flood Insurance Program in stating that all important infrastructure should be above the Base Flood Elevation (BFE). In accordance with the expectation that flooding will worsen in the future, airports should consider elevating charging infrastructure above the BFE, using guidelines like those laid out in the Federal Flood Risk Management Standard. The National Electric Vehicle Infrastructure (NEVI) Formula Program, for example, states that proposed EVSE will be evaluated partly on its consideration of projected future flood plains[52]. For EVSE that cannot be located outside or above floodplains, FEMA and the American Society of Civil Engineers provide in-depth guidelines for flood resistant design standards.

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Americans with Disabilities Act Considerations

Public use infrastructure like charging stations at airports must be planned and constructed to meet the requirements of the ADA. Planning with the ADA in mind ensures that airports are not only following regulations for charging stations, but also helping to expand the general accessibility of EVs.

Alongside state and local laws, the ADA sets minimum enforceable standards that airports must follow when installing charging stations. While the ADA does not provide specific design standards for all aspects of charging stations, it does provide some specific information for the design and layout of charging stations and EV parking spaces that are accessible and usable by people with disabilities.[53] Per guidance from the U.S. Access Board, two key sets of features should be considered for charging stations:

- **Accessible mobility features.** A reasonable number of chargers must have physical access for people who use mobility devices, such as wheelchairs, scooters, walkers, and canes.[54] Accessible mobility features primarily involve having a large enough vehicle charging space, providing access aisles, installing charging infrastructure in accessible locations, and ensuring access to the physical operability of EVSE.
- **Accessible communication features.** All chargers should have accessible communication features and operable parts. This enables chargers to be used by people who are deaf or hard of hearing, little people, and other people with disabilities who do not need accessible mobility features (like access aisles) to use an EVSE port.

A summary of important ADA requirements can be found in Table 7. It is important for airports to recognize that many states and municipalities enforce more stringent accessibility requirements, requiring additional accessible spaces for charging stations and enforcing specific site layout.

Table 7. Summary of Important Americans with Disabilities Act Requirements

Element	ADA/ABA 2004 ANSI A117.1 2003
Number of Spaces	A minimum of 4% of parking spaces in any given lot, or one for every 25 spaces, must be designated as accessible; one out of every six spaces should be van accessible
Parking Stall	8x18 feet for a car and 11x18 feet for a van
Accessible Route Slope/Cross Slope	Maximum 1:20 (5%) running slope and 1:48 (2%) cross slope; Accessible vehicle spaces 1:48 (2%) in all directions and 90-inch clearance for vans
Reach Range	Between 15 and 48 inches on the front and side to allow for reach to all operable parts from a wheelchair
Accessible Controls	Operable with one hand and not requiring grasping, pinching, or twisting of the wrist or force of more than 5 lbs. (the exception is gas pumps)
Accessible Ramps	A ramp or curb-cut must be accessible to allow for operation of the charging station
Facility Accessibility	Must be connected by a minimum of 50-inch-wide accessible route in proximity (not necessarily adjacent) to the entrance of the building
Side Access Aisle	Side access aisle of 60 inches wide to allow space for a wheelchair and equipment in and out of space
Other Considerations	Ensure that bollards, wheel stops, or curb do not obstruct use of the charging station

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Source: The U.S. DOE Alternative Fuels Data Center[55]

What are state and local considerations for charger installations?

Codes

Applicable codes for charging stations are typically governed by national, state, and local electric code and state and local building code. The specific codes an airport must adhere to when planning and constructing charging stations and the degree to which those codes address EVSE will be highly dependent on the jurisdiction the airport falls into. The National Electric Code (NEC), for example, provides requirements for the EVSE location, conductors, and circuit protection, but not all states and municipalities have adopted the entirety of the NEC[56]. Airports and entities planning and constructing charging stations at airports should always seek out the most up-to-date codes, recognizing that states and municipalities may have additional requirements for EVSE.

As the market share of EVs grows, many states and municipalities have introduced EV readiness requirements into their building codes. Depending on jurisdiction, airports may need to ensure a minimum number of chargers and EV-ready parking spaces for both new and existing construction[57]. Common requirements include that 5% of all parking spaces have charging infrastructure installed and between 10% and 40% of spaces be futureproofed for EVs to some degree.

While not always available, utilities can serve as a centralized resource for airports looking to determine the relevant electric and building codes that they must adhere to for charging stations. Guidelines from utilities can include regulations on cords, EV connectors, markings, overcurrent protection, receptacles and power outlets, personnel protection, electricity back-fees, and safety switches.[58] As a part of the utility engagement process detailed in Chapter 6, airports should request resources for relevant charging station codes.

Permitting

Like the codes regulating EVSE, the permitting process for charging stations will be highly dependent on the relevant jurisdiction. Some states like California have been aggressive in encouraging a uniform process for EVSE across municipalities and local utilities provide resources to identify the relevant permitting requirements.[59] Across all jurisdictions, permit applications typically include significant documentation of station design aspects including load calculations, electrical service assessments, site diagrams, and charging layouts. Final site inspection and charger commissioning is also often part of the permitting process.

During permit review, applications can be considered for compliance with electrical, accessibility, building, fire safety, structural and engineering and public safety regulations. In an ideal environment, all aspects of the permit application will be reviewed, and any revisions or comments will be noted and returned in a single package. This allows the permit submitter to respond to all revisions at the same time, which reduces the time and resources required to obtain a site permit. However, as noted in Chapter 5, airports should plan for delays to project timelines as permit application review and revisions are commonly a lengthy and resource-intensive process.

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Charging Station Safety

EVs and charging stations present airports with new and unique safety concerns and involve different safety procedures than other vehicles and fueling stations. High voltage and high current electrical connections and batteries used in EVs, and onsite energy storage systems can be a significant risk to airport personnel, customers, and assets. Because of these safety considerations, it is critical that airports adhere to electric, building, and fire codes, and work with experienced and certified contractors when executing charging station projects. Airports should also ensure that any EVSE is regularly inspected and maintained and that EVSE with connectivity adheres to the relevant communication security protocols.

One of the most important charging station safety and training considerations for airports is fire. Batteries used in EVs, and energy storage systems introduce significantly different fire response considerations than traditional vehicles and fuels. As a result, charging station equipment should be labeled according to fire code, feature electrical disconnects, and verified to be capable of handling EV charging load.[60] Relevant airport staff should be trained on how best to respond to EV and charging station fires, including learning manufacturer-specific EV power shutoff and BEV fire procedures.

Wayfinding and Station Signage Requirements

Signage for charging stations provides two major functions: wayfinding signage helps EV drivers navigate to charging stations, and station signage marks stations and communicates station rules and policies. An example of EV charging signage can be found in Figure 29. Proper signage at charging stations provides the following benefits:[61]

- Helps EV drivers navigate to and identify charging stations.
- Optimizes the use of charging infrastructure by helping all drivers understand that parking spaces at charging stations are only for EVs.
- Provides information about policies—such as access, time limits, and hours of use—and facilitates enforcement.
- Enables the uptake of EVs by providing visibility of charging infrastructure to prospective EV drivers.
- Identifies charging station corridors.

Figure 29. Example of EV Charging Signage



Source: DOE

Minimum standards for signage on public use roads and areas are set by the Federal Highway Administration and published in the Manual on Uniform Traffic Control Devices, which provides approved interim designs for charging station wayfinding signs, as well as general information about placement and visibility.[62]

Station signage designed to enforce policies related to charging must be supported by local ordinances specifying charging station policies. Typically, pavement markings are not enforceable on their own but can be used to reinforce signage. States and municipalities may have additional requirements for signage.

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Chapter 6. Funding Charging Infrastructure

As identified in Chapter 3, there are several different EVSE ownership models that may be implemented based on airport contracting requirements, the use case being served, the operators' appetite for risk and experience with EVs, funding availability, and other factors. The cost for charging infrastructure projects can vary dramatically based on the number and type of charging technologies, which may dictate the level of investment and ownership models that an airport chooses.

EVSE use case will be central to the selection of ownership approach:

- **Public chargers**, such as those serving TNCs and travelers, are expected to have high rates of usage and higher fees for dispensing of electricity than private chargers. Public charges are more likely to be owned and operated by EVSPs or other third parties.
- **Private chargers**, such as those serving eGSE or airport fleet vehicles, are more likely to be owned and operated by an airport due to dedicated vehicle charging with known utilization.

Airports may choose to work with a third party who will own and operate all charging across all use cases. Alternatively, at airport may wish to own all EVSE across the airport to control charger O&M, the setting of fees and increase the revenue generation potential. It is likely that many airports will employ both ownership approaches will be employed, depending on the use case. Ownership models are detailed in *Who owns and operates charging infrastructure?*

What federal funding sources are available?

Airport operators should understand what grants and incentives may be available through federal, state, and utility programs that may reduce the capital costs associated with EV and EVSE procurement and electrical system infrastructure upgrades. Available funding programs may also offer technical assistance that can help with site design, workforce training, utility coordination, and overall project implementation. Many EVSPs and other partners also offer services to help guide customers through incentive programs, which may be of particular interest to small airports with limited financial and labor resources.

Federal Funding

A summary of key federal funding programs that may be applicable to EV and infrastructure deployment at airports can be found in Table 8. A summary of federal tax credits relevant to procurement of EVs and charging infrastructure can be found in

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Table 8. Key Federal Grant Programs Relevant to Electric Vehicle Infrastructure at Airports

Title	Description	Funding Type
Clean Heavy-Duty Vehicle Program	$1 billion for U.S. Environmental Protection Agency (U.S. EPA) program through 2031. $400 million to communities in nonattainment areas. Grants and/or rebates to eligible recipients, like airport sponsors, to replace existing heavy-duty vehicles with clean ZEVs, and for zero-emission infrastructure and workforce development and planning. **Federal Cost Share:** TBD	Competitive
Congestion Mitigation and Air Quality (CMAQ) Improvement Program	$2.5 billion in funding to DOT annually. The CMAQ Program provides funding to state departments of transportation (DOTs), local governments, and transit agencies, including airport sponsors, for transportation projects and programs that help meet the requirements of the Clean Air Act by reducing mobile source emissions and regional congestion on transportation networks. Projects supported with CMAQ funds must demonstrate emissions reductions and be in or benefit a U.S. EPA-designated nonattainment or maintenance area. **Federal Cost Share:** 80%	Competitive
Voluntary Airport Low Emissions (VALE) Program	Allows airport sponsors to use Airport Improvement Program (AIP) funds and Passenger Facility Charges (PFC) to finance low-emission vehicles, refueling and recharging stations, gate electrification, and other airport air quality improvements. An eligible airport must be a commercial service airport that is in a nonattainment or maintenance area for one of the National Ambient Air Quality Standards. **Federal Cost Share:** 75% to 100%	Competitive and Formula
Airport Zero Emissions Vehicle and Infrastructure Pilot Program	Program allows airport sponsors to use Airport Improvement Program (AIP) funds to purchase ZEVs and to construct or modify the infrastructure needed to use such vehicles. Any public-use airport eligible to receive these grants in the National Plan of Integrated Airport Systems is eligible for the program, and airports in U.S. EPA-designated nonattainment areas are given priority for consideration. **Federal Cost Share:** 70% to 100%	Competitive and Formula
Diesel Emissions Reduction Act	Combined $115 million U.S. EPA program grants in FY 2022 and FY 2023 for projects that achieve significant reduction in diesel emissions. Grant funding is available to entities for the retrofit or replacement of existing diesel engines, vehicles, and equipment. **Federal Cost Share:** 45% for vehicle replacement with zero-tailpipe emission power source	Competitive

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Title	Description	Funding Type
Alternative Fuel Infrastructure Tax Credit	Up to 30% of the cost of charging infrastructure, not to exceed $100,000 per site. Qualified fueling equipment must be installed in a location that meets one of three requirements: • The census tract is not an urban area • The population census tract has a poverty rate of at least 20% • The metropolitan and non-metropolitan area census tract has a median family income of less than 80% of the state median family income **Federal Cost Share:** To receive the full 30% credit, projects must meet prevailing wage and apprenticeship requirements. Tax exempt entities can receive a payment equal to the full value of the credit	Tax Incentive

Table 9. Key Federal Tax Incentives for EVs and Charging Infrastructure

Provision	Incentive Value	Details
45W Commercial Fleet EV tax credit	Up to $7,500 for light-duty and up to $40,000 for all other vehicles	• Commercial clean vehicles can claim a credit of 30% of the cost of the vehicle, up to $7,500 in the case of a vehicle that weighs less than 14,000 pounds, and up to $40,000 for all other vehicles. • The tax credit amount also may not exceed the amount by which the EV exceeds the cost of a "comparable" (otherwise comparable in size and use) internal combustion powered vehicle. • Battery size of 15 kWh minimum.
30C Commercial EV Charger Tax Credit	Tax credit covers 6% (up to $100,000 per station) of the cost of the equipment	• Commercial entities can be eligible for a tax credit of up to 30% if the EV charger meeting certain requirements: - Located in population census tract where the poverty rate is >20% - Located in Metropolitan and non-metropolitan area census tract where the median family income is less than 80% of the state medium family income level - Eligible projects must also meet apprenticeships and prevailing wage requirement • This credit cannot be used to cover permitting and inspection fees

There are several requirements for federal grants that airports should consider when planning charging infrastructure projects and pursuing federal monies. During the planning phase, airports should be prepared to undergo the NEPA Review Process for their projects.[63] NEPA requires environmental impact analyses of proposed projects that are federally funded. The environmental review process must be completed before a project commences. However, most projects would likely fall under a categorical exclusion and not require further NEPA consideration. VALE and ZEV funding have traditionally been prioritized to airports located in EPA-designated nonattainment zones.

During the planning phase of a project, airports should also consider placement and purpose of charging infrastructure. For example, many airports operate shuttle buses for transporting passengers and employees around the airport from parking facilities to terminal buildings or other airport facilities. Airports should consider the operation-ownership model they currently have in place or may pursue

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when looking to install charging infrastructure for electric shuttle buses. Airports often contract out shuttle bus services to a third party, and contractors may operate facilities on non-airport-owned property. Grant assurances stipulate that federally financed infrastructure must be built on airport-owned property. Airports looking to install charging infrastructure at contractor-owned bus depots and maintenance facilities would not be eligible for federal funding.

In the procurement phase of a project, there are other applicable requirements tied to federally funded projects. The Davis-Bacon and Related Acts[64] applies to federally funded projects more than $2,000 for construction. Under this act, contactors installing charging infrastructure must pay local prevailing wages, as set by the Department of Labor. Purchased equipment must also meet the provisions of the Build America Buy America Act.[65] This act, which was enacted as part of the IIJA on November 15, 2021, established a domestic content procurement preference for all federal financial assistance obligated for infrastructure projects after May 14, 2022. The domestic content procurement preference requires that all iron, steel, manufactured products, and construction materials used in covered infrastructure projects are produced in the United States. Specifically for FAA funding, the Buy American Preferences[66] requires that all steel and manufactured goods used in Airport Improvement Program (AIP) funded projects be produced in the United States. There are certain instances in which the federal government may waive these Buy American Preferences.[67] These are:

- Applying the Buy American Preferences would be inconsistent with the public interest
- The steel or goods produced in the United States are not produced in a sufficient and reasonably available amount or are not of a satisfactory quality
- The cost of components and subcomponents produced in the United States is more than 60% of the cost of all components of the facility or equipment procured and final assembly will occur in the United States
- Including domestic material will increase the cost of the overall project by more than 25%

The FAA maintains a list of nationwide waivers that have been issued on its website.[68] Airports should work with funding agencies to understand current waivers and to navigate the waiver process for the Buy American requirements applicable to the specific project. Procurement documents should communicate prevailing wages; Buy American requirements; and any other local, state, or federal requirements to prospective vendors and contractors.

Over the course of the project and for a certain time after installation, there are recordkeeping and reporting requirements for federal funding.[69] State funding apparatuses also likely have recordkeeping and reporting requirements. Quarterly and annual reports are typically required over the course of the project. Airport sponsors are required to provide financial and performance reports, including information on purchase and maintenance costs, maintenance records, emissions, and emissions reductions. Airports may be required to keep records pertaining to financial information such as invoices and canceled checks, procurement information such as contracts and purchase orders, administrative information such as grant applications and agreements, and project information such as drawings and reports for up to three years after the final payment. Projects receiving federal funding are also subject to federal audit requirements.

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Several funding mechanisms have been used to replace older diesel equipment, such as the federal Diesel Emissions Reduction Act (DERA) grants, DERA grants administered by the states, and Volkswagen Mitigation funds administered by the states. Some of these grants and funds also require that the diesel equipment be decommissioned, by scrapping the equipment or permanently disabling the engine. Federally funded equipment is also required to be kept on airport property for the duration of its life, with few exceptions, such as vehicle servicing, as outlined by the FAA's ZEV guidance.[70]

State Funding

The availability of funding programs for charging infrastructure at airports varies by state. However, there are several federal programs with state allocations, including state-administered DERA grants and Volkswagen Environmental Mitigation Trust funds. In addition, the Federal Highway Administration is administering the NEVI Program,[71] funded through the 2021 IIJA. The NEVI Program has a state allocation, as well as the federal Formula Program to develop a nationwide network of charging infrastructure. A full list of state-level funding for EV and charging infrastructure is available in several locations such as the DOE's Alternative Fuels Data Center[72] and third-party organizations like Electrek.[73]

Utility Funding

There are significant grant, rebate, and incentive programs available from utilities across the United States. One of the largest and most common types of utility funding are Make Ready programs, in which utilities will pay for some or all the electrical improvements needed to support charging at a property, including (in some programs) BTM components like electrical panels, conduit, trenching, and other items. These incentives typically do not include funds for EVSE hardware. Other utilities may provide specific grants or other incentives for the procurement or installation of EVSE. Examples of utility programs can be found in Table 10. Note that this is not an exhaustive list: a full inventory of utility incentive programs can be found in an online article by Electrek.[74]

Table 10. Examples of Utility Incentive Programs

Utility	Program	Description
Joint Utilities of New York	EV Make Ready Program	May provide up to 100% of the electric infrastructure costs associated with new nonresidential charging stations.
Georgia Power	Electric Transportation Make Ready Program	Designed for EVSE installations that are public facing or part of public fleets serving the public good. Georgia Power will design, install, own, and maintain the BTM electrical equipment up to the charging pedestal.
Southern California Edison	Charge Ready Transport	Southern California Edison (SCE) will design, construct, and install the necessary infrastructure on both the utility side and customer side of the electric meter. SCE will install a separate meter dedicated to the charging infrastructure and waive customer demand charges through 2024 with its commercial EV rates.
Consumers Energy	PowerMIDrive Program	Commercial customers installing qualified, publicly accessible charging stations are eligible for rebates up to $5,000 per Level 2 charging station and up to $70,000 per DCFC station installed.

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Chapter 7. Metrics for Tracking Charger Deployment

This chapter provides context, metrics, and strategies for monitoring EVSE as airports begin to deploy new charging infrastructure. It is important to track not only the number and level of chargers available, but also the performance of charging infrastructure across a variety of measures using data from EVSE, EVSPs, and utilities over time. Effective monitoring of EVSE will significantly improve airports' ability to make informed decisions about future charging infrastructure deployment.

What data should be tracked to monitor the performance of chargers?

Managing charging infrastructure performance saves money and improves user satisfaction. As detailed in Chapter 2, networked chargers typically come with an online portal allowing airports to monitor key metrics. There are several common metrics for tracking EV chargers:[75]

- **Energy usage:** Daily, monthly, or annual energy use is important to track to monitor growth in charging demand or unexpected energy losses (due to malfunction). For a networked charger, the EVSE owner/operator should compare energy use from the charger with that from the electric meter (assuming the charger has its own meter).
- **Charger utilization:** Utilization does not have a standard industry definition but is typically the fraction of time an EVSE port is actively charging an EV out of a 24-hour period. In general, utilization of 20% or more is a high utilization.
- **Idle time:** Vehicle idling is when a vehicle is plugged in but not actively charging. This metric helps inform pricing (if there is a payment system and user fee) and load flexibility (i.e., can the charging be done at a time when electricity is cheaper).
- **Revenue:** The revenue of a charging station refers to the total income generated by the station. There are several possible revenue streams for charging stations, but it is important to keep track of the overall revenue generated by chargers and the entire charging station to make decisions about future charging infrastructure.
- **EVSE uptime:** The uptime of a charging station or an individual charger refers to the availability of charging services in a calendar year. Uptime is a critical metric for assessing the reliability of charging infrastructure. The Federal Highway Administration targets an uptime of 97%.

What are the sources of data that can be used to track performance?

Airports can leverage data from EVSPs and utilities to track performance of charging infrastructure. Each is described briefly below.

- **EVSP Data.** To monitor, control, and visualize charging station data in a central location, management software can provide airports with the ability to view metrics like utilization, revenue, uptime, and energy usage. Management software helps users assess the performance of EV chargers and can provide additional services like payment integration and load management. The management software available to airports will depend on the EVSE provider, whether EV chargers are part of a network of chargers, and whether the EVSE conforms with

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OCPP. Typically, there is some operational cost to use charge management software, and the specific fee structure will depend on the agreement between the airport and the EVSP.

- **Utility Data**. The final source of charging station data is the local utility, which should provide monthly billing data including energy usage, peak load, and fees paid. If installed EVSE is networked, utility billing data can provide airports with the ability to double-check the performance of charging infrastructure. Comparing energy usage and peak load between utility bills and EVSE management software provides the opportunity to identify faulty equipment or inefficient performance. If installed EVSE is non-networked, utility billing data can provide a way to track important EV charging data in lieu of the more advanced metrics provided by charge management software.

Evaluating EVSE performance with utility data can present a challenge. Unlike charge management software, utility billing data will not provide vehicle- or time-specific data, which limits an airport's ability to make informed decisions about when to charge vehicles. This challenge is even more significant if EVSE is not installed on a dedicated meter. Without a dedicated meter, it is very difficult to accurately attribute electricity usage and peak load increase to EVSE.

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Chapter 8. Resiliency, Emergency Management, and Training

What is the impact of EV charging on resiliency planning?

The deployment of EVs and charging infrastructure has both positive and negative impacts on resiliency and emergency response planning. EVs require electricity to charge, which can pose a difficulty during grid failures. Because of the mission-critical nature of airport operations, the FAA requires 50 commercial service airports in the United States to have backup generation that is able to maintain the control tower, airport surveillance radar, approach-light system, instrument landing system, and runway lights on the primary runway for a period of at least four hours.[76] in 2023, the Government Accountability Office (GAO) published a report finding that between 2015 and 2022, 321 commercial service airports reported electrical power outages of five minutes or more, 25% of which lasted for a period of over four hours.[77]

To mitigate the impacts of grid failure events and improve overall energy operations, airports have begun to deploy microgrids, backup generation, renewables, battery storage, and combined heat and power (CHP) systems. Adoption of EVs will have impacts on the total energy demands at airports that should be considered as new technologies are deployed. In addition to overall energy planning, airports can deploy specific technologies at scale specific to EV charging, which are described below.

Backup Generation

Permanent and mobile backup power generation—either diesel-powered or a battery energy storage system—can enhance resiliency of EV charging deployment. Generators present a common and widely used solution for bolstering the resiliency of critical infrastructure when the grid fails. Offering great variety in capacity, mobility, and fuel source, generators can be deployed to sites as needed or can be permanently installed in areas where grid reliability is of great concern or in situations where the infrastructure must always be powered. While modern generators can burn cleaner than their predecessors, they are not an ideal solution for resiliency and sustainability. Generators not only produce significant emissions, but also rely on fuel supply chains that are vulnerable to disruptions, especially during times of increased stress like grid failure.

Figure 30. Fixed Diesel Generator Powering an EV Charger



Source: thedriven.io

Distributed Renewable Generation

Integrating renewable power into new and existing EV charging locations can significantly improve the resilience of charging infrastructure. EV charging can be paired with on-site renewable energy generation, typically on-site solar energy systems, and in some cases batteries, either with or without

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managed charging. This offers the opportunity for emissions-free EV operation, and when paired with energy storage, may enable vehicle operation without grid interconnection.

Solar generation offers flexible location and sizing, zero emissions, and grid independence, which reduces vulnerability and overall demand on the grid. Solar charging can readily support Level 2 EV charging, and adding features like battery storage and grid integration can improve capacity and power delivery. Grid integration also offers the benefit of bolstering overall power generation capability in the area where solar chargers are used.

Energy Storage

The ability to store energy is critical to future grid and charging station resilience. Stored energy not only ensures that stations can continue to charge EVs when grid power fails, it also improves grid stability by providing capacity during peak demand. Integration of energy storage with EV charging is gaining widespread adoption in the United States; for example, Electrify America has deployed battery energy storage systems (BESS) at over 150 EV charging stations across the United States, and deployed its first MW scale installation in 2022. Battery systems can also be used to lower the cost of EV charging at airports by consuming power when electricity costs are lowest (typically overnight) and using it during periods of peak demand. As noted in the section *What is bidirectional charging?* V2X technology may also offer the opportunity for EVs to serve as backup power during events when grid electricity is not available. When pooled together, EVs may be able to delivery electricity to meet airport operational needs during emergency events, such as grid failures, or to help meet demand that cannot otherwise be demand during times of peak consumption. This capacity may also serve to enhance the operations of microgrids at airports in the future if vehicles are parked for consistent periods and have the charging flexibility to charge and discharge while not in use.

Off-Grid Charging

While EVSE installations are primarily dependent upon a grid connection to receive electricity, the deployment of distributed energy resources (DERs) such as solar panels and battery systems can enable EV charging during a grid failure event or without a grid interconnection. Numerous projects throughout the United States pair solar panels with a charging location; however, most of these sites require a grid interconnection for operation. An integrated EVSE + solar + battery storage product can enable an EV charger to be deployed at any location without additional infrastructure requirements (Figure 31).



Figure 31. Beam EV Arc 2020 Off-Grid Charging System

Source: Beam

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Microgrids

At a larger scale, airports are beginning to deploy integrated systems for electricity generation and energy storage that can operate as a grid-connected or stand-alone system to serve their energy demands. Microgrids are groups of interconnected loads and DERs that act as single controllable entities. When disconnected from the main grid, a microgrid can generate, distribute, and store energy for a specific location—such as a hospital, campus, or airport—without a grid connection. For example, in 2021, Pittsburgh International Airport became one of the first airports to have a microgrid, a system that combines a natural gas generator and solar panels to deliver up to 22.5 MW. The airport saved $1 million in energy costs in year one of operations because of this effort.[78]

Microgrid systems are beginning to integrate charging infrastructure in non-airport applications. Planning for EV charging energy demand when designing microgrid systems is important to ensure that capacity is appropriately sized. A more detailed assessment of microgrids for airports can be found in *ACRP Research Report 228: Airport Microgrid Implementation Toolkit.*

What training is required for emergency response?

EVs present specific challenges to emergency responders, particularly in cases of battery fires. EVs are far less likely to catch fire than ICE vehicles; only 337 EV fires have been reported globally since 2010.[79] However, EV fires can be long-lasting and severe, and may require specific fire suppression techniques. EV battery fires can be time- and resource-intensive and can pose safety risks from emission of toxic and flammable gases from damaged batteries and the unpredictability of thermal runaway and reignition.

Flooding, particularly from saltwater, presents specific challenges for the operation of EVs. Residual salt can form conductive bridges that can lead to short circuiting and self-heating of the battery, resulting in fires. The time frame in which a damaged battery can ignite has been observed from days to weeks. While these incidents are extremely rare and safety mechanisms are contained within battery systems, it is still important to follow specific protocols when dealing with EVs damaged by floods.

The National Highway Traffic Safety Administration (NHTSA) has published guidance related to flooding and fires for first responders and second responders, developed in collaboration with the U.S. Fire Administration and National Fire Protection Association (NFPA). The NFPA also provides online and in-person training for first responders for dealing with fires and other situations involving alternative fuel vehicles and technologies.

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Chapter 9. Future Work

The Phase I Planning Guide helped identify key research gaps, which will be the focus of a Phase II project. In addition to an update of this Planning Guide, the Phase II work will address two topics:

PROJECT #1: Calculator for Charging Infrastructure Costs and Electricity Load

Airport planners need better tools to characterize the impacts (e.g., costs, charging needs, and electricity load) of EVs at airports over the next five, ten, or twenty years. The first step in assessing these impacts is to project the number of EVs that will be deployed in the future within major use cases, such as light-duty passenger, medium heavy-duty, GSE, bus, and aircraft segments. From there, with the right data, planners can estimate load impacts and incremental costs associated with charging infrastructure installation, operation, and maintenance. In this project, the Cadmus team will develop an Excel-based decision support Tool (and associated guide) that provides a simple projection of vehicle stock over time and layers in costs, charging needs, loads. Ultimately, the tool and guide will support activities such as:

- Electrical load studies;
- Development of capital and operating budgets for airport carbon reduction targets;
- Determination of federal grant needs; and
- Supporting parking facility planning to provide passenger charging.

PROJECT #2: Guide on Utility Coordination and Energy Master Planning

Electric vehicle charging installation requires utility coordination, particularly for high-powered charging sites needing extensive new electrical equipment. However, airport planners often struggle to understand what information is needed for decision-making by the utility, and accurately assess the timeline for construction and operation of charging projects. There is no current best practice for engaging with utilities to expand electrical infrastructure for EV charging at airports. Furthermore, there is no standard practice for incorporation of EV charging into airport energy master planning. This project will fill these gaps by providing a guide on how to best plan for expanding electrical infrastructure.

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Endnotes

[1] Alliance for Automotive Innovation. 2023. *Electric Vehicle Sales Dashboard.* https://www.autosinnovate.org/EVDashboard

[2] Alternative Fuels Data Center. n.d. *Alternative Fueling Station Locator.* https://afdc.energy.gov/stations/#/analyze?country=US&fuel=ELEC&ev_levels=all&access=public&access=private

[3] The White House. February 25, 2023. "FACT SHEET: Biden-Harris Administration Announces New Standards and Major Progress for a Made-in-America National Network of Electric Vehicle Chargers." https://www.whitehouse.gov/briefing-room/statements-releases/2023/02/15/fact-sheet-biden-harris-administration-announces-new-standards-and-major-progress-for-a-made-in-america-national-network-of-electric-vehicle-chargers/

[4] National Renewable Energy Lab. 2023. "The 2030 National Charging Network: Estimating U.S. Light-Duty Demand for Electric Vehicle Charging Infrastructure." https://www.nrel.gov/news/program/2023/building-the-2030-national-charging-network.html

[5] Alliance for Automotive Innovation. June 26, 2023. "Get Connected: Electric Vehicle Quarterly Report 2023." https://www.autosinnovate.org/EVDashboard

[6] California Air Resources Board. 2023. "Advanced Clean Cars II Regulations: All New Passenger Vehicles Sold in California to be Zero Emissions by 2035." https://ww2.arb.ca.gov/our-work/programs/advanced-clean-cars-program/advanced-clean-cars-ii

[7] California Air Resources Board (CARB). October 2019. "Zero-Emission Airport Shuttle Regulation Factsheet." https://ww2.arb.ca.gov/sites/default/files/2019-10/asb_reg_factsheet.pdf

[8] Le Bris, G., L.-G. Nguyen, B. Tagoe, P. Jonat, C. Y. Justin, E. Reindel, K. B. Preston, and P. J. Ansell. 2022. *ACRP Research Report 236: Preparing Your Airport for Electric Aircraft and Hydrogen Technologies.* Transportation Research Board, Washington, DC. https://doi.org/10.17226/26512

[9] Alternative Fuels Data Center. n.d. "How Do Plug-In Hybrid Electric Cars Work?" https://afdc.energy.gov/vehicles/how-do-plug-in-hybrid-electric-cars-work

[10] Alternative Fuels Data Center. n.d. "How Do All-Electric Cars Work?" https://afdc.energy.gov/vehicles/how-do-all-electric-cars-work

[11] Environmental Defense Fund, April 2023. *Electric Vehicle Market Update. https://www.edf.org/sites/default/files/2023-05/Electric%20Vehicle%20Market%20Update%20April%202023.pdf*

[12] Wood, E., B. Borlaug, M. Moniot, D.-Y. Lee, Y. Ge, F. Yang, and Z. Liu. 2023. *The 2030 National Charging Network: Estimating U.S. Light-Duty Demand for Electric Vehicle Charging Infrastructure.* Golden, CO: National Renewable Energy Laboratory. NREL/TP-5400-85654. https://www.nrel.gov/docs/fy23osti/85654.pdf

Copyright National Academy of Sciences. All rights reserved.

[13] Gabriel, N. January 12, 2023. "$210 Billion of Announced Investments in Electric Vehicle Manufacturing Headed for the U.S." *Atlas EV Hub.*

[14] Alvey, Rebekah, E&E News. October 18, 2023. "EV prices drop as inventory soars, batteries improve." https://subscriber.politicopro.com/article/eenews/2023/10/18/ev-prices-drop-as-inventory-soars-batteries-improve-00120723

[15] Department of Energy. n.d. *All-Electric Vehicles. https://www.fueleconomy.gov/feg/evtech.shtml#:~:text=EVs%20have%20a%20shorter%20driving,take%203%20to%2012%20hours.*

[16] For example, see Southern California Edison's fact sheet on Rule 29: https://www.sce.com/sites/default/files/2022-07/EVRule29FactSheet.pdf

[17] Developing Infrastructure to Charge Electric Vehicles" 2023. U.S. Department of Energy Alternative Fuels Data Center. Accessed September 19, 2023. https://afdc.energy.gov/fuels/electricity_infrastructure.html

[18] Alternative Fuels Data Center. n.d. "Alternative Fueling Station Locator." https://afdc.energy.gov/stations/#/find/nearest

[19] PlugShare. n.d. Mapping site. https://www.plugshare.com/

[20] SAE International. June 27, 2023. "SAE International Announces Standard for NACS Connector, Charging PKI and Infrastructure Reliability." https://www.sae.org/news/press-room/2023/06/sae-international-announces-standard-for-nacs-connector

[21] Le Bris, G., L.-G. Nguyen, B. Tagoe, P. Jonat, C. Y. Justin, E. Reindel, K. B. Preston, and P. J. Ansell. 2022. *ACRP Research Report 236: Preparing Your Airport for Electric Aircraft and Hydrogen Technologies.* Transportation Research Board, Washington, DC. https://doi.org/10.17226/26512

[22] Businesswire. November 7, 2023. "Archer Aviation and BETA Technologies Collaborate to Accelerate the Adoption of an Interoperable Charging System Across the Electric Aviation Industry." https://www.businesswire.com/news/home/20231107548198/en/Archer-Aviation-and-BETA-Technologies-Collaborate-to-Accelerate-the-Adoption-of-an-Interoperable-Charging-System-Across-the-Electric-Aviation-Industry

[23] Plugzio. "Plugzio Outlet Product Description." https://www.plugzio.com/plugzio-integrated-in-wall-outlet

[24] Alternative Fuels Data Center. n.d. "Operation and Maintenance for EV Charging Infrastructure." https://afdc.energy.gov/fuels/electricity_infrastructure_maintenance_and_operation.html

[25] Freewire. "Boost Charger 150." https://freewiretech.com/dc-boost-charger/

[26] Smith, Steve. March 18, 2013. "Electric GSE Buying Trends Report." *Aviation Pros.* https://www.aviationpros.com/gse/gse-technology/green-alternative-energy-gse/article/10889496/electric-ground-support-equipment-buying-trends-report

Copyright National Academy of Sciences. All rights reserved.

[27] Massport. n.d. *Sustainability & Resiliency Report 2020 and 2021.* massport.com/media/bo4gmdk0/2020-2021-sustainability-and-resiliency-report-web.pdf

[28] San Francisco International Airport. n.d. Zero-Emission Vehicle Readiness Roadmap & Intermodal Electrification Strategic Plan: 2020 Executive Summary Report. flysfo.com/sites/default/files/ZEVR_Summary.pdf

[29] Winneshiek Energy District. n.d. "Electrification." Microsoft PowerPoint - 08.PHL Benefits of Equipment Electrification_102417 cleancopy.pptx (dvrpc.org)

[30] Brink, L., et al. (McKinsey & Company). May 31, 2023. "Short haul flying redefined: The promise of regional air mobility." https://www.mckinsey.com/industries/aerospace-and-defense/our-insights/short-haul-flying-redefined-the-promise-of-regional-air-mobility

[31] Mallela, J., P. Wheeler, G. Le Bris, and L.-G. Nguyen. 2023. A*CRP Research Report 243: Urban Air Mobility: An Airport Perspective*. Transportation Research Board, Washington, DC. https://dx.doi.org/10.17226/26899

[32] Federal Aviation Administration. 2023. *Advanced Air Mobility (AAM) Implementation Plan.* https://www.faa.gov/sites/faa.gov/files/AAM-I28-Implementation-Plan.pdf

[33] Le Bris, G., L.-G. Nguyen, B. Tagoe, P. Jonat, C. Y. Justin, E. Reindel, K. B. Preston, and P. J. Ansell. 2022. *ACRP Research Report 236: Preparing Your Airport for Electric Aircraft and Hydrogen Technologies.* Transportation Research Board, Washington, DC. https://doi.org/10.17226/26512

[34] Le Bris, G., L.-G. Nguyen, B. Tagoe, P. Jonat, C. Y. Justin, E. Reindel, K. B. Preston, and P. J. Ansell. 2022. *ACRP Research Report 236: Preparing Your Airport for Electric Aircraft and Hydrogen Technologies.* Transportation Research Board, Washington, DC. https://doi.org/10.17226/26512

[35] Federal Aviation Administration. September 21, 2022. "Engineering Brief 105 Vertiport Design." Memorandum. https://www.faa.gov/sites/faa.gov/files/eb-105-vertiports.pdf

[36] Banchik, L., R. Riedel, and B. Kloss (McKinsey & Company). May 26, 2021. "Right in Your Backyard: Regional Airports are an Accessible and Underused Resource for Future Air Mobility." https://www.mckinsey.com/industries/aerospace-and-defense/our-insights/future-air-mobility-blog/right-in-your-backyard-regional-airports-are-an-accessible-and-underused-resource-for-future-air-mobility

[37] WXY Architecture + Urban Design. November 2012. *Siting and Design Guidelines for Electric Vehicle Supply Equipment*. https://www.transportationandclimate.org/sites/default/files/EV_Siting_and_Design_Guidelines.pdf

[38] SAC County News. August 7, 2018. "County Enters Agreement for EV Charging at SMF." https://www.saccounty.gov/news/latest-news/Pages/County-Enters-Agreement-for-EV-Charging-at-SMF.aspx

[39] Alternative Fuels Data Center. n.d. "Charging Infrastructure Operation and Maintenance." https://afdc.energy.gov/fuels/electricity_infrastructure_maintenance_and_operation.html

Copyright National Academy of Sciences. All rights reserved.

[40] Pacific Gas and Electric. 2019. *Take Charge: A Guidebook to Fleet Electrification and Infrastructure.* CLB-0819-1448. https://www.pge.com/pge_global/common/pdfs/solar-and-vehicles/your-options/clean-vehicles/charging-stations/ev-fleet-program/PGE_EV-Fleet-Guidebook.pdf

[41] Alternative Fuels Data Center. n.d. "Charging Infrastructure Operation and Maintenance." https://afdc.energy.gov/fuels/electricity_infrastructure_maintenance_and_operation.html

[42] Electrify America. n.d. "Pricing and Plans for EV Charging." https://www.electrifyamerica.com/pricing/

[43] Electrify America. n.d. "Pricing and Plans for EV Charging." https://www.electrifyamerica.com/pricing/

[44] McFarlane, Diane, and Matt Prorok. July 31, 2019. *Analysis: How Demand Charges Impact Electric Vehicle Fast Charging Infrastructure.* Great Plains Institute. https://betterenergy.org/blog/demand-charges-and-dcfc/#:~:text=At%2050%20kW%2C%20demand%20charges,81%20percent%20of%20total%20costs

[45] Gnann, T., T.S. Stephens, Z. Lin, P. Plötz, C. Liu, and J. Brokate. October 2018. "What Drives the Market for Plug-In Electric Vehicles? A Review of International PEV Market Diffusion Models." *Renewable and Sustainability Energy Reviews* (93): 158–164. https://www.sciencedirect.com/science/article/pii/S1364032118301497

[46] P.A. Geroski. Models of technology diffusion. Res. Pol., 29 (2000), pp. 603-625, 10.1016/S0048-7333(99)00092-X.

[47] Early deployments of fleet EVs in California suggest ratios of approximately 1 port per EV, although the there are exceptions. https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/sb-350-te/sb-350-standard-review-programs-annual-transportation-electrification-evaluation-2021.pdf

[48] California Public Utilities Commission. June 30, 2022. *Standard Review Projects and AB 1082/1083 Pilots.* https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/sb-350-te/sb-350-standard-review-programs-annual-transportation-electrification-evaluation-2021.pdf

[49] California Air Resources Board. January 2023. "Zero-Emission Airport Shuttle Regulation Factsheet." https://ww2.arb.ca.gov/sites/default/files/2023-01/ASB%20Reg%20Factsheet%20in%20Word_01-12-23.pdf

[50] California Public Utilities Commission. n.d. "Clean Miles Standard." https://www.cpuc.ca.gov/regulatory-services/licensing/transportation-licensing-and-analysis-branch/clean-miles-standard

[51] Mead & Hunt. July 6, 2022. "Charging Infrastructure: Electrifying One Airport at a Time." https://meadhunt.com/charging-infrastructure/

[52] Federal Register, Docket No. FHWA-2022-0008. June 22, 2022. "National Electric Vehicle Infrastructure Formula Program." https://www.federalregister.gov/documents/2022/06/22/2022-12704/national-electric-vehicle-infrastructure-formula-program

Copyright National Academy of Sciences. All rights reserved.

[53] U.S. Access Board. July 17, 2023. *Design Recommendations for Accessible Electric Vehicle Charging Stations.* https://www.access-board.gov/files/usab-evse-guide.pdf

[54] U.S. Access Board. July 17, 2023. *Design Recommendations for Accessible Electric Vehicle Charging: Number of Accessible Chargers.* https://www.access-board.gov/tad/ev/#number-of-accessible-chargers

[55] Alternative Fuels Data Center. n.d. "Installing Electric Vehicle Charging in Compliance with the Americans with Disabilities Act Requirements." https://afdc.energy.gov/fuels/electricity_infrastructure_ada_compliance.html

[56] Energy5. September 30, 2023. "NEC Code for EV Charging Stations." https://energy5.com/nec-code-for-ev-charging-stations

[57] Pacific Northwest National Laboratory. July 2021. *Electric Vehicle Charging for Residential and Commercial Energy Codes: Technical Brief.* https://www.energycodes.gov/sites/default/files/2021-07/TechBrief_EV_Charging_July2021.pdf

[58] Pacific Gas and Electric. *EV Infrastructure Installation Guide*. "Chapter 3. Code Requirements for Installing EVSE." https://www.pge.com/includes/docs/pdfs/about/environment/pge/electricvehicles/ev5pt3.pdf

[59] California Governor's Office of Business and Economic Development. January 2023. *Electric Vehicle Charging Station Permitting Guidebook: Second Edition.* https://business.ca.gov/wp-content/uploads/2019/12/GoBIZ-EVCharging-Guidebook.pdf

[60] National Fire Protection Association. November 28, 2022. "Fire Safety for Electric Vehicles and Other Modern Vehicles in Parking Structures." https://www.nfpa.org/News-and-Research/Publications-and-media/Blogs-Landing-Page/NFPA-Today/Blog-Posts/2022/11/28/EVs-and-Parking-Structures

[61] Alternative Fuels Data Center. n.d. "Signage for Electric Vehicle Charging Stations." https://afdc.energy.gov/fuels/electricity_charging_station_signage.html

[62] Federal Highway Administration. 2009. *Manual on Uniform Traffic Control Devices (MUTCD).* https://mutcd.fhwa.dot.gov/

[63] National Environmental Policy Act. Accessed at https://www.epa.gov/nepa/what-national-environmental-policy-act on October 25,2023.

[64] Davis-Bacon and Related Acts. Accessed at https://www.dol.gov/agencies/whd/government-contracts/construction on October 25, 2023.

[65] Build America Buy America Act. Accessed on October 25, 2023. https://www.hud.gov/sites/dfiles/GC/documents/Build%20America,%20Buy%20America%20Act%20Provisions.pdf

[66] FAA Buy American Preferences. Accessed on October 25, 2023. https://www.faa.gov/airports/aip/buy_american

Copyright National Academy of Sciences. All rights reserved.

[67] Federal Aviation Adminstration. "Buy American Preference Requirements." Accessed October 25, 2023. https://www.faa.gov/airports/aip/buy_american

[68] Nationwide Waivers Issued. Accessed on October 25, 2023. https://www.faa.gov/airports/aip/buy_american/nationwide_waivers_issued

[69] Record Keeping. Accessed on October 25, 2023. https://www.faa.gov/airports/central/airport_compliance/record_keeping

[70] Federal Aviation Administration. Zero Emission Vehicle Guidance. Accessed on October 25, 2023. https://www.faa.gov/sites/faa.gov/files/airports/environmental/zero_emissions_vehicles/Zero-Emissions-Vehicles-Tech-Guidance-v2.pdf

[71] National Electric Vehicle Infrastructure Program. Accessed on November 10, 2023. https://www.fhwa.dot.gov/environment/nevi/

[72] Alternative Fuels Data Center. n.d. "Federal and State Laws and Incentives." https://afdc.energy.gov/laws

[73] Doll, S. June 30, 2023. "Electric Vehicle Tax Credits and Rebates Available in the US, Sorted by State." *Electrek.* https://electrek.co/2023/06/30/ev-tax-credit-rebate-states-electric-vehicles/

[74] Doll, S. June 30, 2023. "Electric Vehicle Tax Credits and Rebates Available in the US, Sorted by State." *Electrek.* https://electrek.co/2023/06/30/ev-tax-credit-rebate-states-electric-vehicles/

[75] EV Charging Summit Blog. 2023. "Top Metrics to Measure the Performance of Your EV Charging Stations." https://evchargingsummit.com/blog/top-metrics-to-measure-the-performance-of-your-ev-charging-stations/

[76] FAA JO 6030.20G https://www.faa.gov/documentLibrary/media/Order/JO_6030_20G.pdf

[77] Government Accountability Office. August 29, 2023. "Airport Infrastructure." https://www.gao.gov/products/gao-23-105203

[78] Wood, E. July 18, 2022. "How a Microgrid Saved Pittsburgh International Airport $1 Million." *Microgrid Knowledge.* https://www.microgridknowledge.com/microgrid-saved-pittsburgh-international-airport-1-million

[79] EV FireSafe. 2023. *Passenger EV Fire Incidents.* https://www.evfiresafe.com/_files/ugd/8b9ad1_4dc7eb412c1e42d487f74c4cc7274e1e.pdf51

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Key Terms Used in this Guide

Battery Electric Vehicle (**BEV):** EV that is powered by an electric motor and that uses a battery that is recharged using an electrical outlet or charging equipment.

Charging Pedestal: The unit that houses all equipment related to charging, such as EVSE ports, user display screen, payment system, and electronic hardware. A pedestal with two EVSE ports can charge two vehicles at the same time.

Charging Station (or station location): Site with one or more charging pedestals at the same address.

Combined Charging System (CSS): Protocol standard that adds pins to the J1772 connector to allow direct current (DC) fast charging. This connector is most commonly supported in North America**.**

Connector: Physical attachment at the end of the charging cable that links the EVSE port to the vehicle. Multiple connector types (e.g., J1772, CCS, CHAdeMO) can be available on a charging pedestal to serve different EV types.

Direct Current Fast Charger (DCFC): Fastest charging level and EVSE type requiring a 480V/3-phase AC power connection with the DCFC equipment converting AC to DC.

Electric Vehicle (EV): Vehicle that uses electrical energy stored in batteries inside the vehicle for propulsion via an electric motor.

Electric Vehicle Supply Equipment (EVSE): Equipment that provides electric power to the vehicle to recharge the vehicle's batteries. EVSE systems include electrical conductors, related equipment, software, and communications protocols that deliver energy efficiently and safely to the vehicle.

Electric Vehicle Supply Equipment (EVSE) Port: Unit that provides power to charge one vehicle at a time. The number of ports on a charging pedestal is equal to the number of vehicles that can simultaneously charge at that pedestal. The piece of infrastructure is commonly referred to as a charger.

Electric Vehicle Service Provider (EVSP): Entity responsible for operating networked or non-networked EVSE, which may include communications systems, billing, maintenance, load management, and other processes. Examples of EVSPs are ChargePoint, Electrify America, and EVgo. Also referred to as Network System Provider (NSP).

Level 1: The slowest EV charging speed and EVSE type requiring 120V 1-phase alternating current (AC) power connection via an electrical outlet.

Level 2: Moderate-speed EV charging level and EVSE type requiring 208V or 240V 1-phase AC power connection via an electrical outlet or hardwired connection.

Light-Duty Vehicle (LDV): Vehicles with a Gross Vehicle Weight Rating (GVWR) of <8,500 lbs., i.e., passenger vehicles.

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Medium- and Heavy-Duty Vehicles (MDHD): Vehicles with a GVWR of >8,500 lbs., i.e., box trucks, buses

Networked EVSE: EVSE ports that have communications systems that enable information to be transferred via Wi-Fi or cellular networks.

Non-networked EVSE: EVSE ports that only deliver power to a vehicle and do not have communications systems.

North American Charging Standard (NACS): Connector that was the proprietary connector for Tesla and is now standardized as the SAE J3400 EV charging system.

Plug-in Hybrid Electric Vehicle (PHEV): EV that is powered by both an internal combustion engine (ICE) and an electric motor and that has a battery that is charged via EVSE that receives power via an electrical outlet or hardwired connection.

SAE J1772: North American standard connector for plugging into EVs, established by the SAE International (formerly the Society of Automotive Engineers). The standard includes physical, electrical, communication, and performance requirements. The J1772 connector is used for both Level 1 and Level 2 charging.

Use Cases: Distinct vehicle-charger combinations defined by their unique operational and charging characteristics.

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